



SKF Bearing greases and their applications

LGMT 2

General purpose industrial and automotive bearing grease

LGMT 2 is mineral oil based lithium soap thickened grease with excellent thermal stability within its operating temperature range. This premium quality, general purposes grease is suitable for a wide range of industrial and automotive applications.

- Excellent oxidation stability
- Good mechanical stability
- Excellent water resistance and rust inhibiting properties

Typical applications:

- Agricultural equipment
- Automotive wheel bearings
- Conveyors
- Small electric motors
- Industrial fans



Bearing operating conditions

Temperature	Medium
Speed	Medium
Load	Low to Medium
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	—
Severe vibrations	+
Shock load or frequent start-up	—
Low noise	—
Low friction	○
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGMT 2

	35 g tube	200 g tube
420 ml cartridge	1 kg can	5 kg can
18 kg can	50 kg drum	180 kg drum

LGMT 3

General purpose industrial and automotive bearing grease

LGMT 3 is mineral oil based lithium soap thickened grease. This premium quality, general purposes grease is suitable for a wide range of industrial and automotive applications.

- Excellent rust inhibiting properties
- High oxidation stability within its recommended temperature range

Typical applications:

- Bearings >100 mm (3,9 in) shaft size
- Outer bearing ring rotation
- Vertical shaft applications
- Continuous high ambient temperatures >35 °C (95 °F)
- Propeller shafts
- Agricultural equipment
- Car, truck and trailer wheel bearings
- Large electric motors



Bearing operating conditions

Temperature	Medium
Speed	Medium
Load	Low to Medium
Vertical shaft	+
Fast outer ring rotation	○
Oscillating movements	—
Severe vibrations	+
Shock load or frequent start-up	—
Low noise	—
Low friction	○
Rust inhibiting properties	○
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGMT 3

420 ml cartridge	1 kg can	5 kg can
18 kg can	50 kg drum	180 kg drum

LGEP 2

High load, extreme pressure bearing grease

LGEP 2 is mineral oil based lithium soap thickened grease with extreme pressure additives. This grease provides good lubrication in operating temperatures ranging from -20 °C (-4 °F) up to 110 °C (230 °F).

- Excellent mechanical stability
- Extremely good corrosion inhibiting properties
- Excellent EP performance

Typical applications:

- Pulp and paper making machines
- Jaw crushers
- Traction motors for rail vehicles
- Dam gates
- Work roll bearings in steel industry
- Heavy machinery, vibrating screens
- Crane wheels, sheaves



Bearing operating conditions

Temperature	Medium
Speed	Low to Medium
Load	High
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	○
Severe vibrations	+
Shock load or frequent start-up	+
Low noise	—
Low friction	—
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGEP 2

420 ml cartridge	1 kg can	5 kg can
18 kg can	50 kg drum	180 kg drum

LGFP 2

Food compatible bearing grease

LGFP 2 is clean, non-toxic bearing grease, which is based on medical white oil using an aluminium complex soap.

This grease is formulated using only FDA* listed ingredients and is authorised by the NSF** for category H1*** service.

- Compliance with all existing legislation on food protection
- High resistance to water washout making it suitable for applications subject to frequent wash down
- Excellent grease life
- Excellent corrosion resistance
- An essentially neutral pH value

Typical applications:

- Bakery equipment
- Food processing equipment
- Multi-pack cassette bearings
- Wrapping machines
- Conveyor bearings
- Bottling machines

- * FDA – Food and Drug Administration
 ** NSF – National Sanitation Foundation
 *** H1 – Incidental Contact with Food



Bearing operating conditions

Temperature	Medium
Speed	Medium
Load	Low to Medium
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	—
Severe vibrations	—
Shock load or frequent start-up	—
Low noise	—
Low friction	○
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGFP 2

SYSTEM 24	420 ml cartridge
1 kg can	18 kg can
180 kg drum	



Re-lubrication



LGFB 2

Food compatible, high load bearing grease

LGFB 2 is PAO/Ester based bearing grease using an aluminum complex soap. LGFB 2 exhibits excellent lubrication ability at high loads and is NSF* H1** certified for use where incidental contact with food cannot be excluded.

- NSF H1 certified
- Resistant to vibration
- High resistance to oxidation
- Shock loads suitability
- Excellent corrosion protection
- High water resistance

Typical applications:

- Any bearing operating under high loads, severe vibrations or shock loads
- Mixers
- Saws
- Slicing machines
- Fillers

* NSF – National Sanitation Foundation

** H1 – Incidental contact with Food



Bearing operating conditions

Temperature	Medium
Speed	Medium
Load	Medium to high
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	○
Severe vibrations	+
Shock load or frequent start-up	+
Low noise	—
Low friction	—
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGFB 2

420 ml cartridge	1 kg can	
18 kg can		180 kg drum

LGEM 2

High viscosity bearing grease with solid lubricants

LGEM 2 is a premium quality, high viscosity, mineral oil based grease using a lithium soap containing molybdenum disulphide and graphite.

- Good lubrication for bearings operating under high loads and slow rotations
- Excellent lubrication properties due to the inclusion of molybdenum disulphide and graphite

Typical applications:

- Rolling element bearings running at low speed and very high loads
- Jaw crushers
- Track laying machines
- Lift mast wheels
- Building machines such as mechanical rams, crane arms and crane hooks



Bearing operating conditions

Temperature	Medium
Speed	Very Low
Load	High to Very High
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	+
Severe vibrations	+
Shock load or frequent start-up	+
Low noise	—
Low friction	—
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGEM 2

SYSTEM 24		
420 ml cartridge	5 kg can	18 kg can
		180 kg drum

LGEV 2

Extremely high viscosity bearing grease with solid lubricants

LGEV 2 is a premium quality, extremely high viscosity, mineral oil based grease using a lithium-calcium soap containing molybdenum disulphide and graphite.

- Excellent lubrication properties due to the inclusion of molybdenum disulphide and graphite solid
- Extremely suitable for lubricating large sized spherical roller bearings subject to high loads and slow rotations, a situation where microslip is likely to occur
- Extremely mechanically stable providing good water resistance and corrosion protection

Typical applications:

- Trunnion bearings on rotating drums
- Support and thrust rollers on rotary kilns and dryers
- Bucket wheel excavators
- Slewing ring bearings
- High pressure roller mills
- Crushers



Bearing operating conditions

Temperature	Medium
Speed	Very Low
Load	High to Very High
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	+
Severe vibrations	+
Shock load or frequent start-up	+
Low noise	—
Low friction	—
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGEV 2

	35 g tube	
420 ml cartridge		5 kg can
18 kg can	50 kg drum	180 kg drum

LGLT 2

Low temperature, extremely high speed bearing grease

LGLT 2 is premium quality, fully synthetic oil based grease using lithium soap. Its unique thickener technology and its low viscosity oil (PAO) provide excellent lubrication performances at low temperatures (–50 °C) and extremely high speeds n.dm values of 1.6×10^6 can be reached.

- Low friction torque
- Low level of power loss
- Quiet running behaviour
- Extremely good oxidation stability and resistance to water

Typical applications:

- Textile spinning spindles
- Machine tool spindles
- Instruments and control equipment
- Small electric motors used in medical and dental equipment
- In-line skates
- Printing cylinders
- Robots



Bearing operating conditions

Temperature	Low to Medium
Speed	Medium to Extremely High
Load	Low
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	—
Severe vibrations	—
Shock load or frequent start-up	○
Low noise	+
Low friction	+
Rust inhibiting properties	○
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGLT 2

	200 g tube	
	1 kg can	
	25 kg can	180 kg drum



Re-lubrication



LGFL 1

Food compatible, low temperature bearing grease

LGFL 1 is PAO/Ester based bearing grease using an aluminum complex soap. LGFL 1 exhibits excellent lubrication ability at low temperature or high speed and is NSF* H1** certified for use where incidental contact with food cannot be excluded.

- NSF H1 certified
- High resistance to oxidation
- Excellent corrosion protection

Typical applications:

- Any bearing operating at low temperature or high speed
- Freezing tunnels
- Conveyors
- Grinders
- Electric motors (small/medium sized-high speed)

* NSF – National Sanitation Foundation

** H1 – Incidental contact with Food



Bearing operating conditions

Temperature	Low to medium
Speed	Medium to extremely high
Load	Low
Vertical shaft	—
Fast outer ring rotation	—
Oscillating movements	—
Severe vibrations	○
Shock load or frequent start-up	—
Low noise	+
Low friction	+
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGFL 1

1 kg can 18 kg can 180 kg drum

LGGB 2

Green biodegradable bearing grease

LGGB 2 is biodegradable, low toxicity, synthetic ester oil based grease using a lithium-calcium thickener. It has excellent lubrication properties for a wide range of applications operating under different conditions.

- Compliance with current regulations on toxicity and biodegradability
- Good performance in applications with steel-on-steel spherical plain bearings, ball bearings and roller bearings
- Good low temperature start-up performance
- Good corrosion inhibiting properties
- Suitable for medium to high loads

Typical applications:

- Agricultural and forestry equipment
- Construction and earthmoving equipment
- Mining and conveying equipment
- Water treatment and irrigation
- Locks, dams, bridges
- Linkages, rod ends
- Other applications where contamination of the environment is a concern



Bearing operating conditions

Temperature	Low to Medium
Speed	Low to Medium
Load	Medium to High
Vertical shaft	○
Fast outer ring rotation	—
Oscillating movements	+
Severe vibrations	+
Shock load or frequent start-up	+
Low noise	—
Low friction	○
Rust inhibiting properties	○
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGGB 2

SYSTEM 24 (LAGD)

420 ml cartridge 5 kg can
18 kg can 180 kg drum

LGWM 1

Extreme pressure low temperature bearing grease

LGWM 1 is a mineral oil based grease using a lithium soap and containing extreme pressure additives. It is extremely suitable for the lubrication of bearings operating under both radial and axial loads e.g. transport screws.

- Good oil film formation at low temperatures down to -30°C (-22°F)
- Good pumpability at low temperature
- Good corrosion protection
- Good water resistance

Typical applications:

- Windmills
- Screw conveyors
- Centralised lubrication systems
- Spherical roller thrust bearing applications



Bearing operating conditions

Temperature	Low to Medium
Speed	Low to Medium
Load	High
Vertical shaft	—
Fast outer ring rotation	—
Oscillating movements	+
Severe vibrations	—
Shock load or frequent start-up	+
Low noise	—
Low friction	—
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGWM 1

420 ml cartridge	5 kg can
50 kg drum	180 kg drum

LGWA 2

High load, extreme pressure, wide temperature range bearing grease

LGWA 2 is premium quality mineral oil based lithium complex grease with extreme pressure (EP) performance. LGWA 2 has such properties that it can be recommended for a wide range of industrial and automotive applications.

- Excellent lubrication at peak temperatures up to 220°C (428°F) for short periods
- Protection of wheel bearings operating under severe conditions
- Effective lubrication in wet conditions
- Good water and corrosion resistance
- Excellent lubrication under high loads and low speeds

Typical applications:

- Wheel bearings in cars, trailers and trucks
- Washing machines
- Electric motors



Bearing operating conditions

Temperature	Medium to High
Speed	Low to Medium
Load	Low to High
Vertical shaft	○
Fast outer ring rotation	○
Oscillating movements	○
Severe vibrations	○
Shock load or frequent start-up	+
Low noise	—
Low friction	○
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGWA 2

SYSTEM 24	35 g tube	200 g tube
420 ml cartridge	1 kg can	5 kg can
	50 kg drum	180 kg drum



Re-lubrication



LGHB 2

High load, high temperature, high viscosity bearing grease

LGHB 2 is a premium quality, high viscosity, mineral oil based grease using the latest complex calcium, sulphonate soap technology. This grease contains no additives and the extreme pressure characteristics are created within the soap structure.

- Excellent anti-oxidation and anti-corrosion properties
- Good EP performance in applications running at high loads

Typical applications:

- Steel on steel plain bearings
- Pulp and paper making machines
- Asphalt vibrating screens
- Continuous casting machines
- Sealed spherical roller bearings operating up to 150 °C (302 °F)
- Withstands peak temperatures of 200 °C (392 °F)
- Work roll bearings in steel industry
- Mast rollers of fork lift trucks



Bearing operating conditions

Temperature	Medium to High
Speed	Very Low to Medium
Load	High to Very High
Vertical shaft	○
Fast outer ring rotation	+
Oscillating movements	+
Severe vibrations	+
Shock load or frequent start-up	+
Low noise	—
Low friction	—
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGHB 2

SYSTEM 24		
420 ml cartridge		5 kg can
18 kg can	50 kg drum	180 kg drum

LGHP 2

High performance, high temperature bearing grease

LGHP 2 is premium quality mineral oil based grease, using a modern Polyurea (di-urea) thickener. It is suitable for ball (and roller) bearings required to run extremely quiet, operating at a wide temperature range from -40 °C (-40 °F) up to 150 °C (302 °F), at medium to high speeds.

- Extremely long life at high temperature
- Wide temperature range
- Excellent corrosion protection
- High thermal stability
- Good low temperature start-up performance
- Compatibility with common Polyurea greases
- Compatibility with lithium complex thickened greases
- Low noise characteristics
- Very good mechanical stability

Typical applications:

- Electric motors: Small, medium and large
- Industrial fans, including high speed fans
- Water pumps
- Rolling bearings in textile, paper processing and drying machines
- Applications with high speed ball bearings operating at medium and high temperatures
- Clutch release bearings
- Kiln trucks and rollers
- Vertical shaft applications
- Vibrating applications



Bearing operating conditions

Temperature	Medium to High
Speed	Medium to High
Load	Low to Medium
Vertical shaft	+
Fast outer ring rotation	—
Oscillating movements	—
Severe vibrations	+
Shock load or frequent start-up	○
Low noise	+
Low friction	○
Rust inhibiting properties	+
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGHP 2

SYSTEM 24		
420 ml cartridge	1 kg can	5 kg can
18 kg can	50 kg drum	

LGET 2

Extreme temperature, extreme condition bearing grease

LGET 2 is premium quality, synthetic fluorinated oil based grease using a PTFE thickener. It has excellent lubrication properties at extremely high temperatures exceeding 200 °C (392 °F) up to 260 °C (500 °F).

- Long life in aggressive environments such as very reactive environments or areas with a presence of high purity gaseous oxygen and hexane
- Excellent oxidation resistance
- Good corrosion resistance
- Excellent water and steam resistance

Typical applications:

- Bakery equipment (ovens)
- Kiln truck wheels
- Load rollers in copying machines
- Wafer baking machines
- Textile dryers
- Film stretching tenders
- Electric motors running at extreme temperatures
- Emergency / hot fans
- Vacuum pumps



Bearing operating conditions

Temperature	Very High
Speed	Low to Medium
Load	High to Very High
Vertical shaft	○
Fast outer ring rotation	+
Oscillating movements	+
Severe vibrations	○
Shock load or frequent start-up	○
Low noise	—
Low friction	—
Rust inhibiting properties	○
+ = Recommended ○ = Suitable — = Not suitable	

Available pack sizes LGET 2

50 g (25 ml) syringe 1 kg can

Important note: Fluorinated greases in general are very costly, however the SKF LGET 2 is competitively priced. Given that LGET 2 is more expensive than other SKF greases, it is therefore recommended to use it only for applications where other SKF greases would not provide the required performance.

SKF Special lubricants



LGRT 2

Railway Axlebox Grease

LGRT 2 is specially developed for lubrication of axlebox bearings of passenger coaches, freight wagons, locomotives, multiple units and mass transit vehicles with speeds up to 200 km/h. It fulfills the European specification EN 12081 (Railway applications - Axleboxes - Lubricating greases) for speed class "a".

- Formulated with a highly refined base oil and lithium soap thickener
- High resistance against oxidation, wear and corrosion due to specially selected additives
- Excellent performance due to high mechanical stability
- Good water resistance
- Does not contain solid ingredients

LDTs 1

Dry Film Lubricant

SKF Dry Film Lubricant LDTs 1 is specially developed for automatic lubrication of flat top chain conveyors in the beverage processing industry. It adheres very well on all treated surfaces and has outstanding properties. The lubricant consists of synthetic oil and is doped with PTFE as solid lubricant. LDTs 1 is NSF* H1** certified for use where incidental contact with food cannot be excluded.

- NSF H1 certified
- Recommended for conveyors using plastic chains
- Excellent lubricating properties

Typical applications:

- Conveyors in bottling lines
- Applications for the following packaging types:
 - Carton packs
 - Cans
 - PET bottles

* NSF – National Sanitation Foundation

** H1 – Incidental contact with Food



SKF SYSTEM 24

Single Point Automatic Lubricator LAGD 60 and LAGD 125

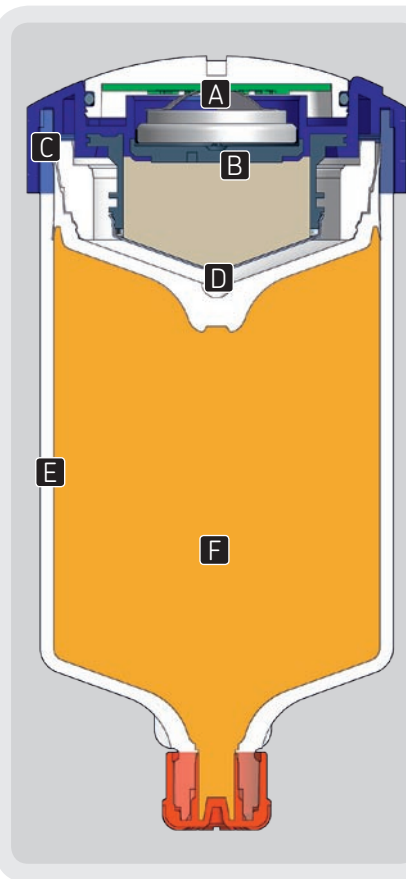
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Reliable, cost effective automatic single point lubricator

The LAGD series of the SKF SYSTEM 24 family are automatic gas driven single point lubricators suitable for many applications. The units are supplied ready-to-use straight from the box and filled with a wide range of high quality SKF lubricants. Tool-free activation and time-setting allows easy and accurate adjustment of lubrication flow.

- Gas driven lubricator is reliable in operation and flexible in application
- Available in two sizes: 125 ml (LAGD 125) and 60 ml (LAGD 60) to suit most bearing lubrication applications
- Compact size, permits installation in restricted areas.
- Available filled with various high quality SKF greases and oils, to suit a wide range of bearing applications
- Flexible time setting period ranging between 1 and 12 months
- Ingress protection to IP 68 level allows the lubricator to be used in most dusty and wet environments
- Wide operating temperature range to suit many applications
- High reliability and dispense rate accuracy helps facilitate predetermined replacement date
- Dispense rate setting is a simple part of the installation process and can be temporarily deactivated
- Transparent lubricant container allows visual inspection of dispense rate
- Intrinsic safety rating: ATEX approved for zone 0
- Wide range of accessories is available
- SKF SYSTEM 24 LAGD series is fully featured in SKF DialSet 4.0





- A Time setting slot**
Allows easy installation and accurate adjustment of lubrication flow
- B Gas cell**
Generates pressure to enable lubricant dispensing
- C Easy-grip top-cover**
Facilitates easy and quick fitting
- D Piston**
Special piston shape helps ensure optimum emptying of lubricator
- E Lubricant container**
Transparent lubricant container allows visual inspection of dispense rate
- F SKF Lubricant**
Filled with a wide range of high quality SKF lubricants

Ordering details

Designation

LAGD 60/WA2
LAGD 125/WA2
LAGD 125/EM2
LAGD 125/GB2
LAGD 125/HB2
LAGD 125/HP2
LAGD 125/FFP2 **

LAGD 60/HMT68 *
LAGD 125/HMT68 *
LAGD 125/HHT26 *
LAGD 125/FHF15 */**
LAGD 125/FU */**

Description

60 ml (2,03 fl oz. US) unit filled with LGWA 2 grease (multi-purpose EP type)
125 ml (4,25 fl oz. US) unit filled with LGWA 2 grease (multi-purpose EP type)
125 ml (4,25 fl oz. US) unit filled with LGEM 2 grease (high loads, slow rotations)
125 ml (4,25 fl oz. US) unit filled with LGGB 2 grease (biodegradable)
125 ml (4,25 fl oz. US) unit filled with LGHB 2 grease (high temperature, loads, plain bearing)
125 ml (4,25 fl oz. US) unit filled with LGHP 2 grease (high performance polyurea)
125 ml (4,25 fl oz. US) unit filled with LGFP 2 grease (food processing industry)

60 ml (2,03 fl oz. US) unit filled with LHMT 68 oil (medium temperature)
125 ml (4,25 fl oz. US) unit filled with LHMT 68 oil (medium temperature)
125 ml (4,25 fl oz. US) unit filled with LHHT 265 oil (high temperature)
125 ml (4,25 fl oz. US) unit filled with LHFP 150 oil (food compatible, NSF H1 approved)
125 ml (4,25 fl oz. US) empty unit suitable for oil filling only

* Includes non-return valve

** Not available in USA and Canada



SKF SYSTEM 24

Single Point Automatic Lubricator LAGE 125 and LAGE 250

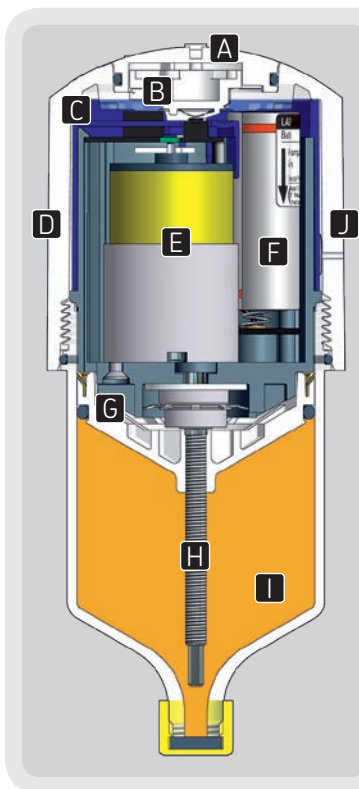


Reliable, reusable lubricator system meets many needs

The LAGE series of the SKF SYSTEM 24 family are single point electro mechanical driven automatic lubricator systems. Suitable for a wide range of applications and operating conditions, the units are reliable and flexible in operation. Supplied ready-to-use straight from the box and ease of installation makes the units a perfect complement to the comprehensive range of SKF automatic lubricators.

- Electro mechanical driven makes the unit highly reliable in operation
- Available in two sizes: 122 ml (LAGE 125) and 250 ml (LAGE 250) to suit most bearing lubrication applications
- Remote mounting up to 3 meters (10 ft) for grease filled units and 5 meters (16 ft) for oil filled units allows lubrication of bearings in areas with high ambient temperatures, vibration or hazardous environments
- Available filled with various high quality SKF greases and oils, to suit a wide range of bearing applications
- Refill sets consisting of a canister filled with SKF grease or oil and a battery pack help ensure reliable lubricator operation
- Flexible user adjustable dispense settings of 1, 3, 6, 9 or 12 months for use in many different applications
- Ingress protection to IP 65 level allows the lubricator to be used in many dusty and wet environments
- Temperature independent dispense rate, suitable for use in applications with changing temperatures
- Unlike gas powered units, a maximum discharge pressure of 5 bar can be achieved over the whole lubricant dispensing period
- Easy activation using a clearly marked dial helps minimize setting errors
- Transparent lubricant container allows visual inspection of dispense rate, whilst electro mechanical functions are indicated by simple red-green LED indicators
- UL listed
- Wide range of accessories is available
- SKF SYSTEM 24 LAGE series is fully featured in SKF DialSet 4.0





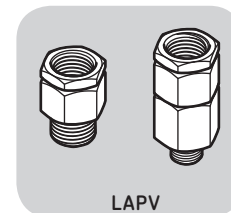
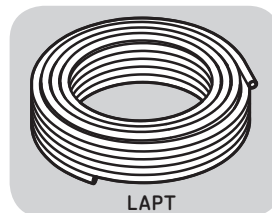
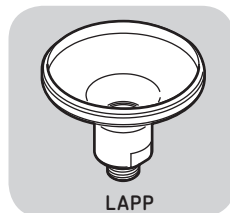
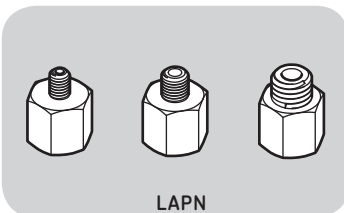
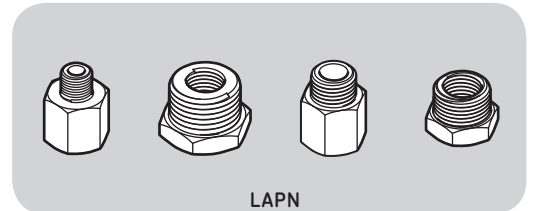
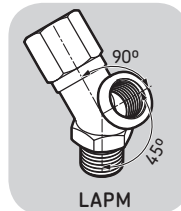
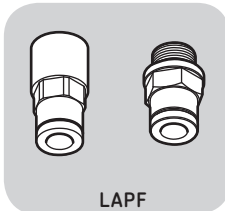
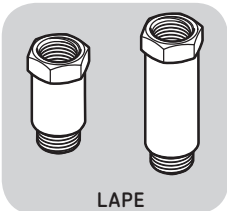
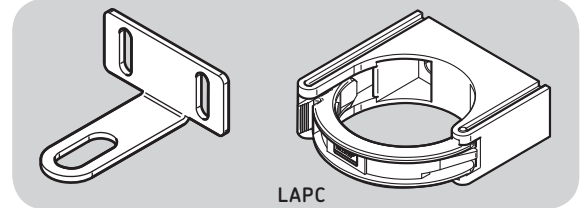
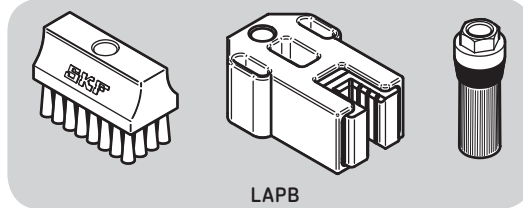
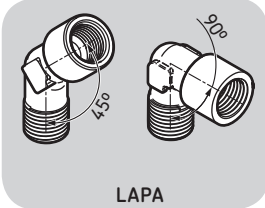
- A B On/off knob and time setting dial**
Enables easy activation and dial setting
- C LED status indicators**
Helps verify operating status
- D Drive cover**
Easily removable; seals and helps prevent ingress of dirt and moisture
- E Electric motor and gearbox**
Helps enable constant discharge pressure
- F Battery pack**
- G Piston**
Special piston shape helps ensure optimum emptying of lubricator
- H Spindle**
Rotates to drive piston, enabling lubricant to be dispensed
- I Lubricant canister**
Filled with high quality SKF lubricant
- J Anti-vacuum membrane**
Helps prevent vacuum forming

Ordering details

Designation	Lubricant	Description	Product
Greases			
LAGE 125/WA2 LAGE 250/WA2 LGWA 2/EML125 LGWA 2/EML250	LGWA 2	Multi-purpose EP type grease	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
LAGE 125/EM2 LAGE 250/EM2 LGEM 2/EML125 LGEM 2/EML250	LGEM 2	High loads, slow rotations	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
LAGE 125/HB2 LAGE 250/HB2 LGHB 2/EML125 LGHB 2/EML250	LGHB 2	High temperature, loads, plain bearing	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
LAGE 125/HP2 LAGE 250/HP2 LGHP 2/EML125 LGHP 2/EML250	LGHP 2	High performance polyurea	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
LAGE 125/FP2 LAGE 250/FP2 LGFP 2/EML125 LGFP 2/EML250	LGFP 2	Food processing industry	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
Oils			
LAGE 125/HMT68 LAGE 250/HMT68 LHMT 68/EML125 LHMT 68/EML250	LHMT 68	Medium temperature oil	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
LAGE 125/HHT26 LAGE 250/HHT26 LHHT 265/EML12 LHHT 265/EML25	LHHT 265	High temperature oil	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250
LAGE 125/HFP15 LAGE 250/HFP15 LHFP 150/EML12 LHFP 150/EML25	LHFP 150	Food compatible, NSF H1 approved oil	Complete unit 125 Complete unit 250 Refill set 125 Refill set 250



Re-lubrication



Accessories ordering details

Designation	Description	Designation	Description
LAPA 45	Angle connection 45°	LAPN 1/8	Nipple G 1/4 – G 1/8
LAPA 90	Angle connection 90°	LAPN 1/2	Nipple G 1/4 – G 1/2
LAPB 3x4E1 *	Brush 30 × 40 mm	LAPN 1/4	Nipple G 1/4 – G 1/4
LAPB 3x7E1 *	Brush 30 × 60 mm	LAPN 1/4UNF	Nipple G 1/4 – 1/4 UNF
LAPB 3x10E1 *	Brush 30 × 100 mm	LAPN 3/8	Nipple G 1/4 – G 3/8
LAPB 5-16E *	Elevator brush, 5 – 16 mm gap	LAPN 6	Nipple G 1/4 – M6
LAPB D2 *	Brush round Ø 20 mm	LAPN 8	Nipple G 1/4 – M8
LAPC 13	Bracket	LAPN 8x1	Nipple G 1/4 – M8 × 1
LAPC 50 **	Clamp	LAPN 10	Nipple G 1/4 – M10
LAPC 63 ***	Clamp	LAPN 10x1	Nipple G 1/4 – M10 × 1
LAPE 35	Extension 35 mm	LAPN 12	Nipple G 1/4 – M12
LAPE 50	Extension 50 mm	LAPN 12x1.5	Nipple G 1/4 – M12 × 1,5
LAPF F1/4	Tube connection female G 1/4	LAPP 2E **	Protection base
LAPF M1/4	Tube connection male G 1/4	LAPP 3E **	Protection cover
LAPF M1/8	Tube connection male G 1/8	LAPP 63 ***	Support flange
LAPF M3/8	Tube connection male G 3/8	LAPP 63V ***	Support flange with non-return valve
LAPG 1/4	Grease nipple G 1/4	LAPT 1000	Flexible tube, 1 000 mm long, 8 × 6 mm
LAPM 2	Y-connection	LAPT 5000 ***	Flexible tube, 5 000 mm long, 8 × 6 mm
LAPM 4 **	Manifold (4 to 1)	LAPV 1/4	Non return valve G 1/4
		LAPV 1/8	Non return valve G 1/8
		LAGE 1-BAT ***	Battery pack

* Suitable for use with oil filled SYSTEM 24 LAGD and LAGE units only

** Suitable for use with LAGD 60 and LAGD 125 SYSTEM 24 only

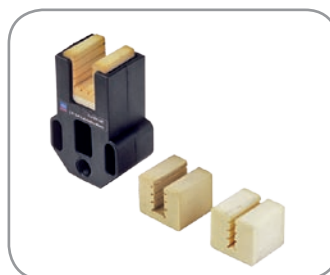
*** Suitable for use with LAGE 125 and LAGE 250 SYSTEM 24 only

Chain oil range

Extending chain life

SKF chain oils come in three convenient sizes to suit the needs of most chain applications in industrial environments. The chain oils - medium temperature, high temperature, and food compatible (NSF H1) - are available in 400 ml (13,52 oz.) aerosol cans, 5 litre (1,32 gallon) cans, and as an oil fill for the SYSTEM 24 single point automatic lubricator.

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Ordering details

Designation	Description
LHFP 150/0.4	400 ml (13,52 oz.) aerosol can
LHFP 150/5	5 litre (1,32 gallon) can
LAGD 125/HFP15 **	125 ml (4,25 fl oz. US) SYSTEM 24 LAGD unit filled with food processing oil (viscosity ISO 150)
LAGE 125/HFP15	122 ml (4,1 fl oz. US) complete unit SYSTEM 24 LAGE filled with food processing oil (viscosity ISO 150)
LAGE 250/HFP15	250 ml (8,5 fl oz. US) complete unit SYSTEM 24 LAGE filled with food processing oil (viscosity ISO 150)
LHFP 150/EML12	122 ml (4,1 fl oz. US) refill set for SYSTEM 24 LAGE filled with food processing oil (viscosity ISO 150)
LHFP 150/EML25	250 ml (8,5 fl oz. US) refill set for SYSTEM 24 LAGE filled with food processing oil (viscosity ISO 150)
LHHT 265/0.4	400 ml (13,52 oz.) aerosol can
LHHT 265/5	5 litre (1,32 gallon) can
LAGD 125/HHT26 *	125 ml (4,25 fl oz. US) SYSTEM 24 LAGD unit filled with synthetic high temperature chain oil (viscosity ISO 265)
LAGE 125/HHT26	122 ml (4,1 fl oz. US) complete unit SYSTEM 24 LAGE filled with synthetic high temperature chain oil (viscosity ISO 265)
LAGE 250/HHT26	250 ml (8,5 fl oz. US) complete unit SYSTEM 24 LAGE filled with synthetic high temperature chain oil (viscosity ISO 265)
LHHT 265/EML12	122 ml (4,1 fl oz. US) refill set for SYSTEM 24 LAGE filled with synthetic high temperature chain oil (viscosity ISO 265)
LHHT 265/EML25	250 ml (8,5 fl oz. US) refill set for SYSTEM 24 LAGE filled with synthetic high temperature chain oil (viscosity ISO 265)
LHMT 68/0.4	400 ml (13,52 oz.) aerosol can
LHMT 68/5	5 litre (1,32 gallon) can
LAGD 125/HMT68 *	125 ml (4,25 fl oz. US) SYSTEM 24 LAGD unit filled with mineral EP type chain oil (viscosity ISO 68)
LAGD 60/HMT68*	60 ml (2,03 fl oz. US) SYSTEM 24 LAGD unit filled with mineral EP type chain oil (viscosity ISO 68)
LAGE 125/HMT68	122 ml (4,1 fl oz. US) complete unit SYSTEM 24 LAGE filled with mineral EP type chain oil (viscosity ISO 68)
LAGE 250/HMT68	250 ml (8,5 fl oz. US) complete unit SYSTEM 24 LAGE filled with mineral EP type chain oil (viscosity ISO 68)
LHMT 68/EML125	122 ml (4,1 fl oz. US) refill set for SYSTEM 24 LAGE filled with mineral EP type chain oil (viscosity ISO 68)
LHMT 68/EML250	250 ml (8,5 fl oz. US) refill set for SYSTEM 24 LAGE filled with mineral EP type chain oil (viscosity ISO 68)

* Includes non-return valve

** Not available in USA and Canada



SYSTEM MultiPoint Automatic Lubricator LAGD 400

Multiple grease lubrication points made easy

The lubrication of bearings with the correct type and quantity of grease is essential for trouble-free operation. Research has shown that 36% of all bearings fail prematurely due to incorrect lubrication. Especially for installations with multiple lubrication points, this can be a time-consuming and costly process. SYSTEM MultiPoint, SKF's centralised automatic lubricator, is the most user-friendly and cost-effective automatic lubricator for multiple grease lubrication points available today. Its compact design, combined with electronically controlled accuracy, makes it an excellent solution for longer bearing-life and increased uptime of your machinery.

Being a do-it-yourself lubrication system, SYSTEM MultiPoint can be easily installed without the assistance of a costly lubrication service company and requires no special training to use. Once the correct grease dispense rate for your application is calculated using DialSet 4.0, SKF's re-lubrication calculation program, SYSTEM MultiPoint will keep up to eight lubrication points simultaneously and automatically lubricated, preventing both over and under-greasing. The transparent cartridge housing allows for easy inspection, while an electronic alarm will warn you when the grease cartridge is empty.

- Do-it-yourself centralised lubrication system
- Up to 8 feed lines
- Easy-to-use
- DialSet 4.0 included: SKF's re-lubrication calculation program allowing accurate calculation of the correct re-lubrication intervals
- Long feed lines (maximum up to 5 m / 16 ft)
- Electronic setting and read-out of control parameters
- Alarm function for blocked feed lines and empty cartridge
- Machine steering (i.e. lubricator only operates while machine is running)
- High-pressure capability (40 bar / 600 psi)
- Tested and approved with all SKF greases
- Uses standard SKF grease cartridges (420 ml)
- Ready for use, all accessories included

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SYSTEM MultiPoint Automatic Lubricator LAGD 1000 series



Centralised lubrication for up to 20 lubrication points

The SKF SYSTEM MultiPoint automatic lubricator LAGD 1000 is a reliable centralised lubrication system offering lubrication solutions for bearings and machinery. The LAGD 1000 uses a high-pressure pump and a progressive feeder for lubricating from 6 to 20 lubrication points.

Supplied as a complete ready-to-use kit, all accessories required are included in the box.

- Suitable for greases from NLGI 000 to NLGI 2, offers high flexibility in types of lubricants that can be used
- Sturdy, robust design with IP65 rating, suitable for most industrial environments
- Transparent reservoir, allows visual monitoring of lubricant level
- Progressive divider so that exactly the same amount of grease reaches each lubrication point
- 1 litre (33,8 US fl. oz) refillable grease container, increases intervals between re-filling
- Extensive programming options, allowing flexibility to suit most applications
- Empty container alarm, helps reduce the risks associated with lack of lubrication
- Wide temperature range, helps meet a variety of tough operating conditions, year round
- Cycle switch function (not on battery version) notifies operator of blocked outlets so that proper action can be taken
- Battery powered version available, no need for external power source, can be used most anywhere
- LAGD 1000 is fully featured in SKF DialSet 4.0





Re-lubrication Calculation Program DialSet 4.0

Accurate calculation of re-lubrication intervals

NEW

DialSet is a calculation program, which easily calculates re-lubrication interval settings. After selecting the criteria and grease relevant to your application, the program provides you with the correct settings for your SKF automatic lubricators.

- Selecting the operating conditions of your application, vertical shaft, outer ring rotation and shock loads, allows accurate calculation of the re-lubrication intervals
- Calculations are based on SKF lubrication theories
- Calculated lubrication interval depends on the properties of the selected grease, minimising the risk of under or over-lubrication and optimising grease consumption
- Calculations are based on SKF automatic lubrication systems grease dispense rates, allowing the recommendation of the correct lubricator setting
- Recommended grease quantity depends on the grease replenishment position; side or W33 for optimum grease consumption
- Includes a complete list of the SKF SYSTEM 24 family accessories

DialSet 4.0 on CD-ROM

DialSet 4.0 is available on CD-ROM with calculation in 10 languages: English, French, German, Italian, Spanish, Swedish, Portuguese, Russian, Chinese and Thai.

The program is suitable for PC's working with MS Windows 98 and later. This CD-ROM can be ordered from SKF under designation MP3506.

DialSet 4.0 online

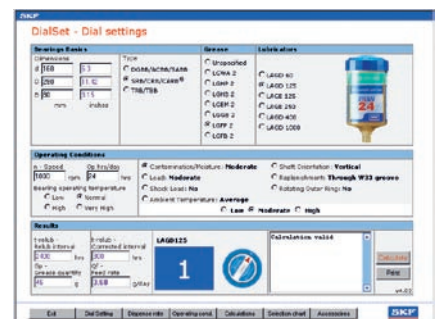
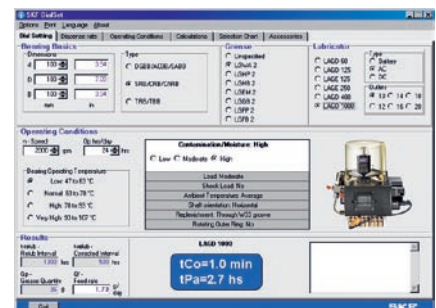
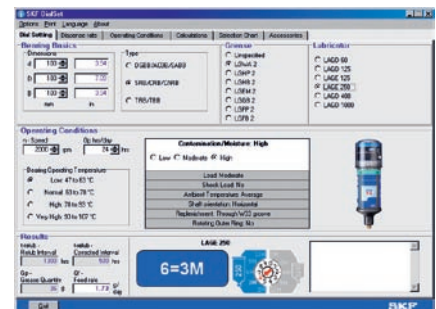
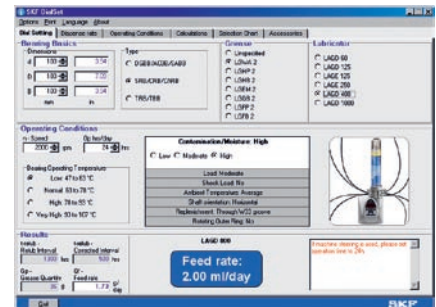
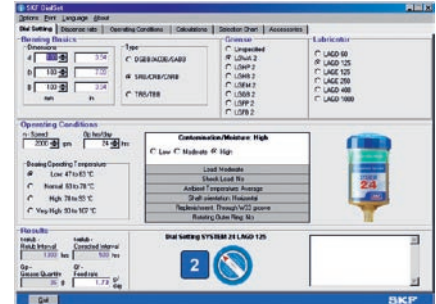
In addition to the downloadable PDA/PPC and the CD-ROM versions, SKF also offers you DialSet 4.0 online in English language.

The program is accessible free-of-charge from www.mapro.skf.com.

After filling in your application's conditions, calculations are made online and the program provides you with a printable re-lubrication interval recommendation.

DialSet 4.0 for PDA/PPC

If you own a PDA or a PPC, you can now calculate the correct re-lubrication intervals on-site. From www.mapro.skf.com, you can now download, free-of-charge, the PDA/PPC version of SKF's re-lubrication calculation program DialSet 4.0 in English language.



Oil Leveller LAHD series

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Automatic adjustment for optimal oil lubrication level

SKF Oil Levellers, LAHD 500 and LAHD 1000, are designed for automatic adjustment of the optimal oil lubrication level within a bearing housing, gear box, crank case or similar oil bath lubrication application. SKF Oil Levellers allow you to effectively adjust the correct oil level during running conditions, optimising machine performance and increasing the service life of the applications.

- Optimally maintained oil level provides adequate lubrication
- Easy visual inspection
- Extended re-lubrication intervals. LAHD 1000 compensates for evaporation losses of up to 1 litre of lubricating oil!
- Oil must be refilled manually

Furthermore, they automatically compensate for oil leakage and offer the possibility of visual inspection of the oil level.



Oil Safe drums and dispensing lids LAOS series

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Help reduce contamination and decrease operating costs

The Oil Safe range of products are ideal for the storage and administration of fluids and oil lubricants. Oil Safe consists of various size drums and dispensing lids, which are fully interchangeable to meet the customer's specific lubrication requirements. The lids are available in ten different colours to help in creating a colour coded system.

- The unique 'O' ring sealing, twist spout and lid design, keep rain, dust and other contaminants out. This helps result in a reduction in the damage caused to plant machinery by the use of contaminated lubricants
- Containers are heat and chemical resistant. They contain UV radiation stabilisers and anti-static agents. The products are durable and the contents are protected from environmental damage even in the harshest of conditions
- Manufactured from HDPE (High Density Polyethylene) grade polymers, helps prevent rust and makes product suitable for indoor and outdoor use
- Lids come in ten different colours allowing different colours to be used for different oils. Colour coded labels are also available to identify the contents
- Drums all have wide mouth openings which allows for fast, non-spill filling from bulk sources

There are five different lids (three with incorporated spout), five different size drums ranging from 1,5 to 10 litres (1,6 to 10.5 quart). In addition, there are two different spout hose attachments and one pump that can be used with the Oil Safe® utility lid.

- Drums are compact (1½, 2, 3, 5 and 10 litre), and ergonomically designed. Makes the product easier to lift and carry
- Drums are see-through and have measuring scales so that fluids and oil lubricant levels are easy to measure and contaminants can easily be seen





Re-lubrication



Grease Packer LAGP 400

To lubricate open bearings

The grease packer LAGP 400 is a low-pressure alternative for emptying SKF grease cartridges. It provides an easy and clean alternative to manual grease packing of open bearings.

- Supplied with three spout caps
- Applies grease to open applications such as bearings or open gears



Grease Gun 1077600

Easy grease filling

The SKF grease gun is ideal for the agricultural, industrial and construction industries and for private use. The SKF grease gun is delivered with a 175 mm (6,9 in) long extension pipe with hydraulic gripping nozzle. A flexible 500 mm (19,7 in) long pressure hose with hydraulic gripping nozzle is available as an accessory.

- For use with cartridges and loose grease
- Rigid hinging system offers long-lasting use
- Knurled body for firm and safe grip
- High quality steel is dent-resistant for easy cartridge loading
- Special piston design for smooth emptying of cartridges
- 40 MPa (5 800 psi) maximum pressure
- 1,5 cm³ (0,092 in³) volume/stroke
- Also available with a 300 mm (12 inch) high pressure hose with a hydraulic gripping nozzle, 1077600H
- A complete set, including 3 extension pipes, high pressure hose, packed in a carrying case is also supplied



Ordering details

Designation

1077600
1077600H
1077601
1077600/SET

Description

Grease gun with extension pipe
Grease gun with flexible hose
Flexible hose
Grease gun set

One Hand Operated Grease Gun LAGH 400

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Easy grease filling with one hand

Suitable for grease filling by grease filler pumps and also suitable for grease cartridges. Ergonomic design, flexible hose and possibility to mount the hose in both vertical and horizontal position make it easy to use.

- Easy-to-use: only one hand is needed to operate the gun
- Refillable: grease filling nipple and de-airing valve allow filling up by filler or grease pump
- Heavy duty: operating pressure up to 30 MPa
- 0,8 cm³ (0,05 in³) volume/stroke
- Flexible hydraulic type hose: can be bent, can be mounted both horizontally and vertically on the gun



Battery Driven Grease Gun LAGG 400B

Quick and easy grease filling

The Battery Driven Grease Gun LAGG 400B is a high quality grease gun suitable for lubricating bearings, machines, vehicles and other applications. Ergonomically designed and user friendly, the grease gun can be used with standard SKF grease cartridges (420 ml) or filled with approximately 500 cm³ (17 US.fl.oz) of loose grease.

- Compared to hand operated grease guns, a 420 ml cartridge can be emptied in approx 10 minutes with minimum effort, resulting in significant cost and time saving
- Electrical operation and the ergonomic design of hand grip helps reduce operator fatigue compared to manual methods
- Easy-to-use one hand operation makes the grease gun user friendly
- Supplied with various electrical plugs and two power supply versions (230 and 110 V charger) makes it operational worldwide
- Battery operation allows use in almost all environments without use of a mains supply
- Long battery life of 1 000 charging cycles helps reduce product life costs
- Refillable: grease filling nipple and de-airing valve allow filling up by filler or grease pump
- Safety valve setting set to 40 MPa (5 800 psi) helps increase operator safety
- Supplied with M10x1 nozzle, interchangeable with SKF hoses, SKF grease meter LAGM 1000E and other accessories



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Re-lubrication



Bearing Packer VKN 550

Contamination free grease filling

The SKF Bearing Packer, VKN 550, is a sturdy, easy-to-use, efficient and effective bearing grease packer. It can also be used in combination with a standard grease gun, air-operated grease pump or grease filler pump. Although specially designed for taper roller bearings, the SKF Bearing Packer works for any type of open bearing which needs to be 100% pre-filled with grease.

- Flushes the grease between the rolling elements where it matters most, prolonging the bearing service life
- Closed system and the cover lid prevent ingress of dirt virtually eliminating contamination
- Allows the operator to pre-fill bearings with grease in a quick and clean way
- Prevents unnecessary grease loss
- Economical and environmentally friendly



Disposable Grease Resistant Gloves TMBA G11D

Skin protection when handling grease

Specially designed to protect the skin when working with SKF bearing grease. The gloves are packed in a handy box containing 50 pairs.

- Non-powdered nitrile rubber gloves
- Close fitting for precision wear
- Excellent resistance against bearing greases
- Non-allergic



Grease Meter LAGM 1000E

Accurate grease quantity measurement for adequate lubrication

It is generally difficult to determine the correct quantity of grease when manually lubricating bearings, either using a grease gun or pump, which can result in either over- or under-greasing the bearing. That can negatively influence the bearing's service life and possibly result in machine breakdown.

The SKF Grease Meter LAGM 1000E accurately measures grease discharge in volume or weight, in both metric (cm^3 or g) and US units (US fl. oz or oz). It has a high maximum pressure of 70 MPa (10 000 psi), making it ideal for use in combination with many types of grease guns and pumps.

- Measures grease discharge in volume or weight, making conversion calculations unnecessary
- High accuracy facilitates adequate bearing lubrication, reducing the risk of over- or under-greasing
- Suitable for all SKF bearing greases of consistency classes up to NLGI 3
- An oil and grease resistant rubber sleeve protects the electronics in case of impact
- The backlit LCD displays large and clear-to-read digits, including "low battery" indication
- Small, compact and lightweight design - only 0,3 kg (0,66 lb)
- Corrosion-free aluminium housing
- Easy to install and use

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Grease Filler Pumps LAGF series

High quantity grease packer

SKF Grease Filler Pumps are suitable for filling grease guns. Especially designed for use on grease gun 1077600 and LAGH 400. Tested and approved for SKF greases. Easy to install and ready for use. Available for standard SKF 18 and 50 kg (39 and 110 lb) drums.

- Quick filling: low pressure allows higher stroke volume
- Easy to install: all necessary items are included
- Reliable: tested and approved for all SKF greases
- Can be used in combination with SKF Bearing Packer VKN 550



Ordering details

Designation	Description
LAGF 18	Grease filler pump for 18 kg drums
LAGF 50	Grease filler pump for 50 kg drums

Grease Pumps LAGG series

Meeting all your grease lubricator needs

Full range of manual and air-operated grease pumps are designed to empty standard 18, 50 or 180 kg (39, 110 or 400 lb) grease drums. Can be connected directly on the greasing points, also suitable for centralised lubricating grease systems.

- Full range; pumps available for 18, 50 or 180 kg (39, 110 or 400 lb) drums
- High pressure; maximum of 42 MPa (6 090 psi)
- Easy to install; all necessary items as well as 3 500 mm (137,8 in) of tubing are included
- Reliable; tested and approved for SKF greases
- Can be used in combination with SKF bearing packer VKN 550

SKF grease pumps have a maximum pressure of 40 and 42 MPa (5 800 and 6 090 psi) respectively. Tested and approved for SKF greases. Easy to install and ready for use since pumps are supplied with all necessary items including 3 500 mm (137,8 in) of tubing.



Ordering details

Designation	Description
LAGG 18M	Grease pump for 18 kg drums
LAGG 18AE	Mobile grease pump for 18 kg drums
LAGG 50AE	Grease pump for 50 kg drums
LAGG 180AE	Grease pump for 180 kg drums
LAGT 180	Trolley for drums up to 200 kg





Re-lubrication



Lubrication accessory sets

Grease Nozzles LAGS 8 / Grease Nipples LAGN 120

The right tools for adequate (re)lubrication

The SKF LAGS 8 Grease nozzle kit provides the user with practical accessories for daily lubrication such as connectors, couplings and nozzles most widely used in the industry.

To meet all of your needs for grease lubrication points, SKF has developed a grease fitting kit, LAGN 120,

which contains a full range of 120 standardised conical grease fittings made of precision steel, zinc plated, hardened and blue chromated.

- Includes the most widely used accessories in the industry
- Upgrade the 1077600 grease gun with the LAGS 8 Grease nozzle kit
- Replace damaged grease fittings

Contents

Designation: LAGN 120

Grease nipple

Quantity

M6x1	straight	30×
M8x1	straight	20×
M10x1	straight	10×
G 1/8	straight	10×
M6x1	45°	5×
M8x1	45°	10×
M10x1	45°	5×
G1/8	45°	5×
M6x1	90°	5×
M8x1	90°	10×
M10x1	90°	5×
G1/8	90°	5×

Contents

Designation LAGS 8

Straight pipe 180 mm and nozzle / Hose / Tube / Tube with nose piece and plastic transparent cover / Nipple M10x1-G1/8 / Nipple M10x1-1/8-27NPS / Nozzle (2×)

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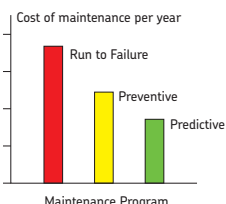
Basic Condition Monitoring

Basic condition monitoring is essential for achieving maximum bearing service life

To help ensure long bearing service life, it is important to determine the condition of machinery and bearings while in operation. Good predictive maintenance will help reduce machine downtime and decrease overall maintenance costs. To help you achieve the maximum service life from your bearings, SKF has developed a wide range of measuring instruments for analysing the critical environmental conditions, which have an impact on bearing and machine performance.

The SKF range covers the most important parameters for measuring machine condition to achieve optimum bearing performance:

- Temperature
- Speed
- Noise
- Oil condition
- Vibrations
- Bearing condition



A



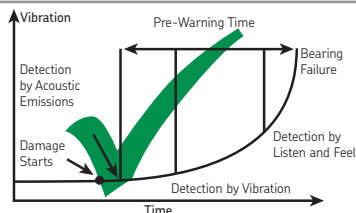
The most expensive maintenance alternative. Maintenance cost comparisons.



B



Preventive maintenance is similar to the regular service of a car. Often unnecessary maintenance is performed.



C

Condition based maintenance means repairs are only carried out when required. The most effective alternative.

Maintenance concepts

A Run to failure

Run to failure occurs when repair action is not taken until a problem results in machine failure. Run to failure problems often cause costly secondary damage along with unplanned downtime and maintenance costs.

B Preventive maintenance

Preventive maintenance implies that a machine, or parts of a machine, are overhauled on a regular basis regardless of the condition of the parts. While preferable to run to failure maintenance, preventive maintenance is costly because of excessive downtime from unnecessary overhauls and the cost of replacing good parts along with worn parts.

C Predictive maintenance

Condition monitoring/predictive maintenance is the process of determining the condition of machinery while in operation. This enables the repair of problem components prior to failure. Condition monitoring not only helps plant personnel reduce the possibility of catastrophic failure, but also allows them to order parts in advance, schedule manpower, and plan other repairs during the downtime.

With condition monitoring, machinery analysis takes two overlapping forms: predictive and diagnostic.

General-purpose Thermometer Pen TMTP 200

Accurate temperature measurement in general industries

The SKF General-purpose Thermometer Pen is a user-friendly, durable pocket size thermometer. Its sturdy flexible probe tip provides effective surface contact for accurate temperature measurement. Since no maintenance engineer should work without one, the thermometer pen is supplied with a handy pouch with belt clip for protection and portability.

- Compact, ergonomic design
- Wide measurement range, from -40 to 200 °C (-40 to 392 °F)
- Temperature reading selection in °C or °F
- Flexible probe tip for better surface contact, providing high measuring accuracy
- Dust tight and water resistant, rated IP 65
- Maximum temperature function allows temperature peak hold
- Auto power off function
- Ultra low power consumption



Non-contact Thermometer TMTL 500

Measuring temperature at a safe distance

Lightweight and compact, the Non-contact Thermometer utilises advanced class II laser beam for accurate aiming and an infrared detector for measuring temperature. The TMTL 500 is extremely user-friendly – simply aim, pull the trigger and read the temperature on the large backlit display. No contact with hot surfaces or moving parts means safer, faster and easier temperature measurement.

- Safely measures the temperature of hot, hazardous or hard-to-reach objects
- Wide measurement range from -60 to 500 °C (-76 to 932 °F)
- Distance/spot ratio of 11:1, better for applications where accuracy is critical
- Ideal complement to other condition monitoring techniques, such as vibration analysis
- Backlit display, making temperature easier to read in dimly lit or dark places
- Temperature reading selection in °C or °F
- Low power consumption using 2x AAA batteries, more energy efficient
- Auto shut off feature, to optimise battery life
- Robust construction for use in an industrial environment



Additionally, the TMTL 500 allows temperature measurement where contact with a conventional temperature probe should be avoided to prevent surface contamination, making ideal to use for measuring temperature of food processing applications.



Basic Condition Monitoring

Infrared Thermometer CMAC 4200-SL

Measuring wide temperature range at safe distance

The SKF Infrared Thermometer CMAC 4200-SL with laser sighting is a rugged, easy to use portable non-contact thermometer. Ideal for a broad range of maintenance tasks, the CMAC 4200-SL may be connected directly to SKF's portable data collectors for quick, accurate recording of temperatures.

- Wide temperature range of $-30\text{ }^{\circ}\text{C}$ to $900\text{ }^{\circ}\text{C}$ (-25 to $1\ 600\text{ }^{\circ}\text{F}$)
- Accuracy $\pm 1\%$ of reading
- Compatible with SKF portable data collectors
- Designed for physically demanding environments
- Easy to use
- 16-Dot laser sighting circle with Distance to Spot size (D:S) of 60:1
- Adjustable emissivity with on-board table
- Fast response time

A useful on-board table including 30 pre-set material emissivity listings is easily accessed, allowing accurate temperature measurement of different material surfaces.



Advanced Infrared and Contact Thermometer TMTL 1400K

Versatility in temperature measurement

The TMTL 1400K combines the flexibility of an infrared thermometer with the facility of a contact thermometer. An object's temperature can be measured using an infrared detector or a probe, making it ideal for situations where accurate temperature measurement is necessary and the emissivity of an object is unknown. It is supplied with a K type probe, has variable emissivity, and many different possible measurement modes.

- User selectable variable emissivity between 0.1 and 1.0, when used in conjunction with the probe the correct emissivity can be defined helping to ensure accurate temperature measurement
- The SKF temperature probe TMDT 2-30 is included (max. $900\text{ }^{\circ}\text{C}$ / $1\ 652\text{ }^{\circ}\text{F}$), ideal for measuring objects with a high temperature
- Safely measures the temperature of hot, hazardous or hard-to-reach objects
- Wide measurement range using infrared sensor from -60 to $500\text{ }^{\circ}\text{C}$ (-76 to $932\text{ }^{\circ}\text{F}$), with probe -64 to $1\ 400\text{ }^{\circ}\text{C}$ (-83 to $1\ 999\text{ }^{\circ}\text{F}$)
- Distance/spot ratio of 11:1, better for applications where accuracy is critical
- Ideal complement to other condition monitoring techniques, such as vibration analysis
- Backlit display, making temperature easier to read in dimly lit or dark places
- Temperature reading selection in $^{\circ}\text{C}$ or $^{\circ}\text{F}$
- Low power consumption using 2x AAA batteries, more energy efficient
- Auto shut off feature that can be programmed from 1 minute to 1 hour, to optimise battery life
- Robust construction for use in an industrial environment



Dual Laser Infrared Thermometer TMTL 2400K

Advanced infrared and contact temperature measurement



The SKF Dual Laser Infrared Thermometer TMTL 2400K can be used as both a contact and non-contact thermometer. The object's temperature can be measured using an infrared detector or a probe.

Lightweight and compact, the TMTL 2400K can safely measure the temperature of hot, hazardous or hard-to-reach objects without contaminating or damaging the object's surface. The optics of the TMTL 2400K sense emitted, reflected and transmitted energy, which is collected and focused onto a detector.

- Can be used as infrared thermometer and/or contact thermometer with supplied K type temperature probe (compatible with other probe types available from SKF)
- Non-contact measurement makes it possible to safely measure the temperature of hot, hazardous or hard-to-reach applications
- Wide measurement range using infrared sensor from -60 to 1 000 °C (-76 to 1 832 °F), with probe -64 to 1 400 °C (-83 to 1 999 °F)
- Distance/spot ratio of 50:1, suitable for applications where accuracy is critical
- User selectable variable emissivity between 0,1 and 1,0
- White LED Flashlight to help visualize the application
- Large backlit display enables easier temperature readings in dimly lit or dark places

The TMTL 2400K's electronics translate the collected information into a temperature reading, which is displayed in °C or °F. The built-in dual-laser feature and spot ratio of 50:1 helps enable exact pinpointing of the area of interest.

The TMTL 2400K enables contact temperature measurement and allows the emissivity to be user selected. This allows the contact temperature measurement to be easily compared with the infrared measurement in order to find and adjust the thermometer emissivity for accurate infrared measurement of the same measured object.



K-type Thermocouple Probes TMDT 2 series

Wide variety of thermocouple probes for many applications

SKF offers fifteen K-type thermocouple probes for use with the SKF digital thermometer TMTL 1400K and TMTL 2400K.



Typical applications are:

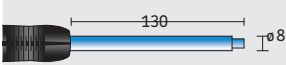
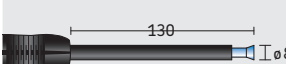
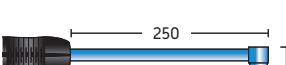
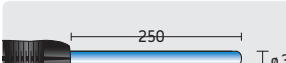
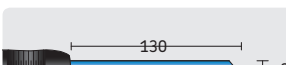
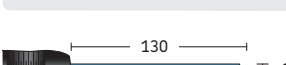
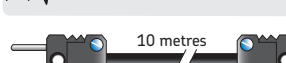
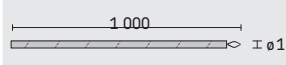
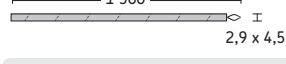
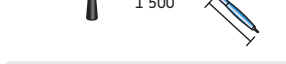
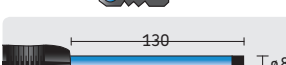
- Surface measurements (TMDT 2-30, -31, -32, -33)
- Gas and liquid measurements (TMDT 2-34)
- Semi solid materials (TMDT 2-35)
- Clamp for pipe measurements (TMDT 2-36)
- Rotating surface measurements (TMDT 2-40)
- Liquid non-ferrous metal measurements (TMDT 2-41)
- Ambient temperature measurements (TMDT 2-42)
- Gas measurements – wire probes (TMDT 2-38, -39)
- Heavy-duty surface measurements (TMDT 2-43)

All probes can be used with the SKF digital thermometers TMTL 1400K and TMTL 2400K without recalibration.



Basic Condition Monitoring

K-type thermocouple probes

Designation	Description	Dimensions (mm)	Max. temp	Response time
TMDT 2-30	Standard surface probe For hard surfaces such as bearings, bearing housings, engine blocks, oven shields, etc.		900 °C 1 650 °F	2,3 sec
TMDT 2-31	Magnetic surface probe For hard, magnetic surfaces; the integral heat sink design and low mass minimise thermal inertia and provide an accurate temperature measurement.		240 °C 460 °F	7,0 sec.
TMDT 2-32	Insulated surface probe For hard surfaces where electrical wiring might cause short circuiting, e.g. electric motors, transformers, etc.		200 °C 390 °F	2,3 sec.
TMDT 2-33	Right angle surface probe For hard surfaces in heavy-duty applications, e.g. machine components, engines, etc.		450 °C 840 °F	8,0 sec.
TMDT 2-34	Gas and liquid probe Flexible shank made of stainless steel for liquids, oils, acids, etc. and at high temperature, e.g. open fire (not for molten metals).		1 100 °C 2 010 °F	12,0 sec.
TMDT 2-34/1.5	Gas and liquid probe Same as TMDT 2-34 but with thin shank and faster response time. Very flexible, specially suitable for measuring temperature of gases.		900 °C 1 650 °F	6,0 sec.
TMDT 2-35	Probe with sharp tip Can be easily inserted into semi-solid materials like food-stuffs, meat, plastic, asphalt, deep-frozen products, etc.		600 °C 1 110 °F	12,0 sec.
TMDT 2-35/1.5	Probe with sharp tip Same as TMDT 2-35 but with thinner shank and faster response time for insertion into soft solids.		600 °C 1 110 °F	6,0 sec.
TMDT 2-36	Pipe clamp probe For temperature measuring on pipes, cables, etc. Diameter up to ø 35 mm (1,4 in).		200 °C 390 °F	8,0 sec.
TMDT 2-37	Extension cable For use with all K-type probes. Special lengths are available on request.			
TMDT 2-38	Wire probe Thin, light weight, very fast response, fibreglass insulated.		300 °C 570 °F	5,0 sec.
TMDT 2-39	High temperature wire probe Thin, light weight, very fast response, ceramic insulation.		1 350 °C 2 460 °F	6,0 sec.
TMDT 2-40	Rotating probe For moving or rotating smooth surfaces. Four roller bearings provide suitable contact with the surfaces. Max. velocity 500 m/min.		200 °C 390 °F	0,6 sec.
TMDT 2-41	Non-ferrous foundry probe Holder including dip-element for molten, non-ferrous metals. Highly resistant to corrosion and oxidation at high temperatures.		1 260 °C 2 300 °F	30,0 sec.
TMDT 2-41A	Dip-element Replacement dip-element for TMDT 2-41.		1 260 °C 2 300 °F	30,0 sec.
TMDT 2-42	Ambient temperature probe For measurement of ambient temperature.			
TMDT 2-43	Heavy duty surface probe Same as TMDT 2-30 but with silicon encapsulated tip for heavy duty applications.		300 °C 570 °F	3,0 sec.

Advanced Thermal Imager TMTI 2

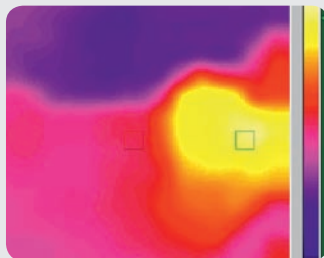
Comprehensive high resolution thermal imaging package



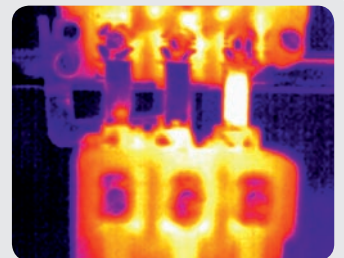
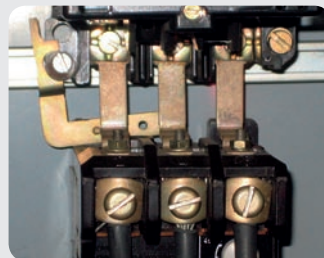
The Advanced Thermal Imager TMTI 2 is a user-friendly thermal imager, producing high quality images from invisible infrared radiation. The high resolution (160x120 /19 200 pixels) images are displayed on a large backlit 3½” screen, for ease of viewing. Powerful PC and report-writing software is supplied as standard (MS windows 2000, XP or Vista required), which help enable the user to perform comprehensive data analysis and enhancement of the images.

- Compact ergonomic design and light weight helps make the TMTI 2 suitable for use in most industrial applications
- Easy to navigate menu makes the TMTI 2 a user-friendly instrument and helps facilitate ease of use
- Non-contact measurement technique enabling measurements to be made safely on running equipment
- Wide temperature measurement range from -10 °C to 250 °C is ideal for Predictive Maintenance applications
- User selectable emissivity correction from 0,2 to 1,0 with reflective ambient temperature correction, helps ensure that the TMTI 2 is suitable for almost any application
- Convenient temperature measurement in K, °C and °F eliminates the necessity to convert temperature reading
- Laser pointer pinpoints the area of interest. Easy to associate the hot spot on the screen with the target in the field
- 4 different selectable colour pallets (ironbow, rainbow, high contrast rainbow and greyscale) for ease of viewing
- 20° x 15° field of view easily enables the application to be seen
- Large thermal image storage capacity, up to 1 000 images (in radiometric format) can be stored on the SD card provided. Easy for data collection and subsequent reporting
- Up to 10 live images per minute can be displayed on a PC

Advanced temperature measurement features are built into the TMTI 2 such as, automatic hot spot and cold spot finder, Maximum/Minimum temperature view of an area and temperature difference display of two user defined spots. Suitable for industrial environments, the TMTI 2 is supplied in a heavy duty case with software, rechargeable battery, battery charger, and a SD memory card for image storage.



The TMTI 300 (see next page) unit is used to show a difference in temperature between two running bearings. This temperature difference could indicate a potential bearing problem that could lead to a failure and downtime.



The TMTI 2 unit is used to inspect cable connections. The temperature of one of the cable connections is significantly higher than the others. This could indicate potential problems and should be further investigated.



Basic Condition Monitoring



Thermal Imager TMTI 300

Affordable thermal imaging for effective maintenance

The SKF TMTI 300 is a flexible, easy-to-use thermal imager that produces visible images from invisible infrared radiation. It enables effective viewing of a wide range of temperatures for safe inspection of mechanical and electrical machinery.

- Easy-to-use, light weight, one or two handed operation allows the TMTI 300 to be flexible and used for most industrial applications
- Non-contact measurement technique enables measurements to be made safely on running equipment
- Large thermal image storage capacity, 1 000 images per Mb can be stored on Pocket PC/ storage card. Easy for data collection and subsequent reporting
- Two user defined spot temperature measurement points allow comparison of areas of interest. The temperature difference of the 2 spots is displayed as a separate value
- Both PC and "Pocket PC" compatible allowing flexibility in viewing results and report writing
- Software included for ease of data analysis
- Laser pointer shows the size of a pixel, allowing the area of interest to be pinpointed and for accurate measurement
- Convenient temperature measurement in K, °C and °F eliminates the necessity to convert temperature reading
- 3 different selectable colour pallets (green/blue, red/blue, greyscale) for ease of viewing
- Tripod mounting thread for stability and stable monitoring over a period of time
- Sturdy ready-to-use carrying case. The imager, pistol grip and pocket PC can be stored as one assembly ready for use
- Ideal complement to other condition monitoring techniques, such as vibration analysis

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Sound Pressure Meter TMSP 1

Easy noise level measurement

The SKF TMSP 1 is a high quality, hand-held instrument for measuring the sound level in decibels. The environmental noise is picked up by the electret condenser microphone and then processed by the handset. The surrounding noise can be monitored both quantitatively and qualitatively.

The SKF Sound pressure meter TMSP 1 is supplied in a carrying case complete with windshield, calibration screwdriver, jack for external outputs and an alkaline battery.

NEW

- User friendly and easy to operate, no special training required
- dBA and dBC scale weightings for both general sound level and low frequency noise measurements suits most applications
- Fast and Slow time weighting selection to enable either normal measurements or measurement of average level of fluctuating noise
- Four different measurement scales to suit almost all eventualities
- Wide operating temperature suitable for diverse industrial environments
- User selectable backlight for use in environments with poor lighting
- Four digit LCD panel with both digital and bar graph display
- Max and min function for peak measurements
- Alarm functions (Under/Over) to indicate when the noise level is too low or too high
- Tripod mounting facility for use when the instrument must remain in the same position for a prolonged period
- Conforms to IEC651 type2, ANSI S1.4 type 2 for sound level meters

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Multi-functional Laser / Contact Tachometers TMRT series

Pinpoint accuracy combined with measurement versatility

The SKF TMRT series includes two user-friendly and accurate tachometers utilising laser or contact for measuring rotational and linear speed: TMRT 1 and TMRT 1Ex. Equipped with laser and contact adaptor, both tachometers offer excellent speed measurement versatility in five different modes.

Additionally, their large angular range of $\pm 80^\circ$ to target facilitates the easy measurement in areas where straight-line access is difficult. The laser optical system allows easy and quick speed measurement at safe distance from rotating machinery.

Intrinsically Safe Tachometer TMRT 1Ex

The SKF TMRT 1 is also available in an intrinsically safe (Ex) version, especially designed for use in potentially explosive hazardous areas. The TMRT 1Ex has been tested and certified according to the latest ATEX standards in intrinsic safety zones generally found in industries such as the petrochemical, gas and pharmaceutical among others. EC Type Examination Certificate Baseefa03ATEX0425X.

II 2 G EEx ia IIC T4

- The user can select to measure:
 - rpm, rps, m, ft or yds per minute or second,
 - length or revolution counting, or
 - time interval
- Wide speed range and the various measurement modes make the TMRT series suitable for measuring speed in many applications
- Large angular range of $\pm 80^\circ$ to target facilitates easy measuring in areas where straight-line access is difficult
- The large inverting LCD display facilitates easy reading even when pointing the unit down into the machinery
- Compact design; one-hand operated instrument
- Supplied in carrying case for protection and portability
- The TMRT 1 can also be equipped with remote laser sensor, which is optionally available



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Basic Condition Monitoring

Stroboscope TMRS 1

Easy, cost effective inspection in a flash

The SKF TMRS 1 is a portable, easy-to-use stroboscope that allows the motion of rotating or reciprocating machinery to appear frozen, facilitating inspection without stopping the machine.

- The bright flash allows better illumination of the application at a distance, giving a wider viewing area.
- Flash rates of up to 12 500 flashes per minute (FPM) cover a wide range of applications
- Flash rate is quick and easy to adjust using the variable dial rate. Allowing the required speed to be reached within a matter of seconds
- Phase shift mode for optimum inspection of gears, rolls, fans, pulleys. The feature of interest can be rotated to the correct position for inspection
- $\times 2$, $\div 2$ buttons for quick adjustment of FPM
- Easy to read LCD display
- Compact design, one-hand operated instrument
- Battery powered with long running time per charge (up to 2,5 hrs)
- Includes universal AC adaptor that can be used worldwide
- Extra flashtube supplied to minimise downtime of unit
- Supplied in carrying case for protection and portability
- Mounting thread on the underside allows mounting on a tripod for stability and ease of use

Equipped with a phase shift feature that allows the user to advance or retard the flash timing without changing the flash rate, the motion can be “frozen” at the position required for inspection.



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Endoscope TMES series

Easy, cost effective inspection in restricted spaces

The SKF TMES series are compact, lightweight endoscopes that can be used for effective visual inspection of machinery in restricted spaces. The SKF TMES series includes two endoscopes:

- Compact, lightweight design makes the endoscope really portable
- High quality optics allow good image resolution for diagnostic purposes
- Long fibre optic tube with small diameter enables access almost anywhere
- Fully flexible fibre optic tube with a 40 mm (1.6 in) bending radius allows use in applications with tight corners
- Water resistant flexible tube can be used in applications where moisture is present
- Built-in battery powered light source can be adjusted to help prevent over illumination of the application
- Length of handle can be reduced for use in areas with limited space
- Straight view allows the image at the tip to be directly seen without having to make time-consuming adjustments
- Wide field of view offers an excellent viewing range for restricted space applications
- Easy to assemble and use, thus special training is not required
- Digital camera adapter is available as an accessory in order to record results for report writing or sending image on for diagnostics

TMES 1 (Basic) and TMES 2 (Advanced). Equipped with a built-in variable light source and a long flexible fibre optic tube they are suited to many industrial applications.

TMES with optional digital camera adaptor. Camera not included



Not for medical applications



Basic Condition Monitoring



Electronic Stethoscope TMST 3

Easily pinpoints bearing and machine noise

The SKF TMST 3 is a high quality, sensitive instrument enabling the determination of troublesome machine parts by the detection of machine noises or vibrations. TMST 3 includes a headset, two different length probes (70 and 220 mm) and a pre recorded audio CD demonstrating the most common encountered troublesome machine noise, all supplied complete in a sturdy carrying case.

- Excellent sound quality helps to reliably identify the possible cause of the noise
- User friendly and easy to operate, no special training required
- Lightweight ergonomic design makes it easy to operate with one hand
- Excellent quality headset for optimum sound quality even in very high-noise environments
- Pre-recorded demonstration CD and output for analogue recording help facilitate analysis and comparison
- Supplied with two probes, 70 and 220 mm (2,8 and 8,7 in) enabling reach almost anywhere
- Adjustable digital volume control up to 32 levels to reach desired volume

and a pre recorded audio CD demonstrating the most common encountered troublesome machine noise, all supplied complete in a sturdy carrying case.

NEW

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Ultrasonic Leak Detector TMSU 1

Quick and easy detection of air leaks

The SKF TMSU 1 is a high quality, user friendly, sensitive instrument enabling the detection of air leaks by means of ultrasound. Leaks are caused by fluid flowing from a high pressure environment to a low pressure environment, creating turbulence. The turbulence generates high frequency sounds (so called ultrasound) that can be detected by TMSU 1.

- Lightweight compact design makes it easy to operate with one hand
- User friendly and easy to operate, no special training required
- Highly reliable, helps identify exact position of the leak
- Valuable instrument for enhancing machinery sustainability. By identifying gas leaks and fixing them, energy consumption is significantly reduced
- Sensor mounted on a flexible tube helps enable reach in restricted areas
- Excellent quality headset for optimum sound quality even in very high-noise environments, also act as ear defenders
- Wide operating temperature suitable for diverse industrial environments

The sensitive piezoelectric crystal in the sensor allows the operator to quickly guide the instrument to the loudest point which helps locate the leak location. TMSU 1 includes an ultrasonic detector, headset, rubber nozzle and batteries, all supplied complete in a sturdy carrying case.

*Note:
The TMSU 1 is not ATEX approved*

NEW

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OilCheck Monitor TMEH 1

Detects changes in oil condition

The OilCheck measures the changes in dielectric constant of an oil. By comparing the measurements obtained from used and unused oils of the same type and brand, the SKF OilCheck is able to determine the degree of change in the condition of the oil. Dielectric change is directly related to the degradation

and the contamination level of the oil and will allow the user to achieve optimised intervals between oil changes and detect increased mechanical wear and loss of the oil's lubricating properties. To facilitate trending the instrument is equipped with a numerical read-out.

Note

The SKF OilCheck is not an analytical instrument. It is an instrument to detect only changes in the oil condition. The visual and numerical read-outs are purely a guide to enable trending of the comparative readings of a good oil to a used oil of the same type and brand. Do not rely solely on numerical readings.

- Shows changes in oil condition affected by such things as:
 - Water content
 - Fuel contamination
 - Metallic content
 - Oxidation
- Hand held and user friendly
- Numerical read-out to facilitate trending



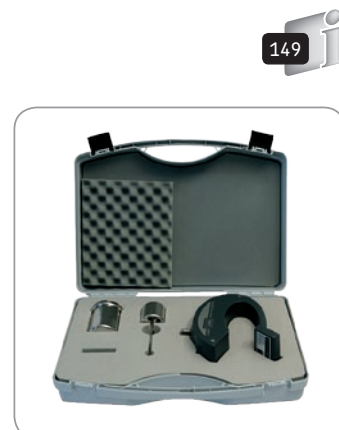
Handheld Viscometer TMVM 1

Reduce costly and time consuming laboratory analyses

The SKF viscometer TMVM 1 is a handheld rotational viscometer for quick and reliable viscosity measurements of lubrication oils and hydraulic fluids. It is suitable for both quantitative and qualitative measurements.

The TMVM 1 is a first check device for carrying out on-site measurements. Regular viscosity checks help provide timely information regarding oil condition, which can have an effect on lubrication and machine performance.

- Compact light-weight design and portability of the viscometer make it an essential device for field service engineers and other field operators
- Supplied standard with a rotor suitable for a viscosity range between 30 and 1 300 mPas covering the viscosity range relating to the greater majority of lubrication oils
- Ergonomic design makes it easy to operate with one hand
- Direct read out eliminates the need for further complex laborious calculations
- Use of supplied measuring cup helps ensure reproducible measuring conditions for accurate measurements
- Excellent repeatability allowing reliable trending of oil condition
- Recalibration message displayed on the viscometer allows timely recalibration
- Alternative rotors available as accessories make viscosity measurement of other fluids possible
- Battery (4 × AA alkaline) operated, does not require a mains electrical supply; therefore a convenient instrument for field tests
- Supplied in a sturdy carrying case for ease of portability





Basic Condition Monitoring

Inspector 400 Ultrasonic Probe CMIN 400-K

Easy detection of high frequency sounds

The Inspector 400 Ultrasonic Probe senses high frequency sounds produced by operating equipment, leaks and electrical discharges. It electronically translates these signals by a heterodyning process, making them audible,

so that a user can hear these sounds through a headset and see them as intensity increments on a meter.

- Detects pressure and vacuum leaks, including compressed air
- Checks steam traps and valves quickly and accurately
- Detects arcing, tracking and corona in electric apparatus
- Tests bearings, pumps, motors and compressors
- Frequency response: 20 - 100 kHz (centred at 38 - 42 kHz)
- Indicator: 10-segment LED bar graph (red)



Vibration Pen^{plus} CMVP 40 and CMVP 50

A powerful combination for detection of machine and bearing defects

A multi-parameter approach to condition monitoring provides two different methods for monitoring machinery condition. This allows early detection of specific machinery problems and provides more ways to measure changes in machinery condition. The Vibration Pen^{plus} is a multi-parameter vibration-monitoring tool capable of measuring vibration caused by rotational and structural problems such as unbalance,

misalignment and looseness. It is also capable of measuring vibration in higher frequencies caused by rolling element bearing or gear mesh problems. This approach provides accurate and reliable data upon which to base maintenance decisions and promotes early detection, confirmation and accurate trending of bearing and machinery problems.

- Measures ISO 10816, low frequency vibration, from 10 Hz to 1 kHz, for overall machine condition
- Assess vibration in industrial non-reciprocating machinery
- Acceleration enveloping for early warning of bearing and gear mesh faults
- So light and compact it fits in your shirt pocket
- Easy one button operation
- Easy to read dual value display



Basic Condition Monitoring Package CMPK series

Check bearing and machine condition quickly and easily

The CMPK series is an essential collection of monitoring tools that no industrial manufacturing plant should be without. These tools make condition monitoring a simple task for maintenance, operations, reliability and vibration analysis departments.

Kit CMPK 200^{plus} (Metric) includes:

- Vibration Pen^{plus} with Carrying Case CMVP 50 (mm/s, RMS – Metric)
- Inspector 400 Ultrasonic Probe with Headphones CMIN 400
- Non-contact MicroTemp Thermometer CMSS 2020
- Batteries included
- VibCard
- Belt Holder for the Vibration Pen^{plus}
- Comprehensive Quick Start Instruction Card
- Rugged Carrying Case

Kit CMPK 210^{plus} (English) includes:

- Vibration Pen^{plus} with Carrying Case CMVP 40 (in/s, eq. peak – English)
- Remainder of items same as the CMPK 200^{plus} package



Bearing Analysis Kit CMPK series

Check bearing and machine condition quickly and easily

The Bearing Analysis Kit is a convenient collection of monitoring tools that no industrial manufacturing plant should be without. It makes condition monitoring a simple task for maintenance, operations, reliability and vibration analysis departments.

Bearing Analysis Kit CMPK 60^{plus} (English) includes:

- Vibration Pen^{plus} CMVP 40 (in/s, eq. peak – English), including manual, carrying case, severity card and battery
- Laser Sighted Non-Contact Temperature Probe CMSS 2000-SL, including manual, hard case, belt clip and battery
- OilCheck Monitor TMEH 1, including soft carrying case and battery
- Custom hard-shell carrying case

Bearing Analysis Kit CMPK 70^{plus} (Metric) includes:

- Vibration Pen^{plus} CMVP 50 (mm/s, RMS – Metric), including manual, carrying case, severity card and battery
- Remainder of items same as the CMPK 60^{plus} kit





Basic Condition Monitoring

MicroVibe P CMVL 3850

Analysis power without complexity

This economical vibration meter expansion module fits in a Pocket PC's compact flash card slot (CF Type II) and features the user-friendly Windows Mobile Operating System. Identify problems and assess machine condition quickly and easily with this versatile and easy-to-use pocket tool.

- Universal PDA platform with user-friendly Windows™ Mobile OS
- Displays overall vibration, time-waveform, FFT spectrum analysis and early indication of bearing degradation
- Easily operated by novice and experienced users
- On-board vibration dictionary
- Enables experienced Pocket PC users to upload overall scalar and spectral data to PC for trending and further analysis with included Data Management Software
- Kit includes MicroVibe P Module, MicroVibe P Data Management Software, accelerometer and cable, stinger and magnet, earphones and carrying case (does not include PDA)

The MicroVibe P collects and displays overall vibration readings and automatically provides expert judgment of the measured velocity and overall enveloped acceleration levels, enabling immediate, accurate and reliable assessment of machine or bearing condition.



MARLIN Condition Detector Pro IS CMVL 3600-IS

Intrinsically safe automatic collection of vibration and temperature data

The MARLIN Condition Detector Pro IS (MCD) is certified as Intrinsically Safe (IS) for use in the hazardous environments typically found in the Petrochemical Industrial marketplace.

The sensor of the MARLIN Condition Detector Pro IS affixes to a machine point via a MARLIN QuickConnect (MQC) Stud or magnetic bases for automatic collection of vibration and temperature data. Green, yellow and red LEDs provide easy to interpret indications of equipment status, so operations or maintenance personnel can quickly identify the need for more in-depth analysis on a particular machine.

The MARLIN Condition Detector Pro IS operates as a stand-alone device, or as an integral component of the complete MARLIN System. By pairing the MARLIN Condition Detector Pro IS with the MARLIN data managers (MDM), important machinery and process information may be stored for trending review and detailed analysis.

- Vibration velocity and enveloped acceleration (gE), and temperature measurements
- Use with MARLIN, ODR data collectors or stand alone
- Red, Yellow, and Green LED alarms for quick go-no-go indication of equipment status
- Interfaces with the MARLIN QuickConnect (MQC) mechanical/computerized studs provides for a fast, quarter turn connection which temporarily fastens the probe to a measurement point
- Intrinsic Safety (IS): LCIE:
 - Approved to CENELEC EN50 020, EEX ia lic T4
 - CSA: Class I, Division 1, Groups A, B, C, D T3A (USA, Canada)





Dismounting

Bearing dismounting

Mechanical dismounting

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Bearing dismounting

Reduce the risk of damaging components and personal injury

When dismounting bearings, care must be taken not to damage other machine components, such as the shaft or housing, as damage can result in compromising the machine's efficiency and lifetime. Bearings are sometimes dismounted to maintain or replace other components of the machine. These bearings are often re-used. Selecting the correct dismounting methods and tools is then essential in reducing the risk of damaging the bearing, allowing it to be used again. Dismounting bearings can be a hazardous and demanding task. Selecting the correct dismounting methods and tools is therefore of utmost importance for reducing the risk of personal injuries. Individual applications may require mechanical, heat or hydraulic dismounting methods and tools to allow safe, correct and efficient bearing dismounting.

Mechanical dismounting

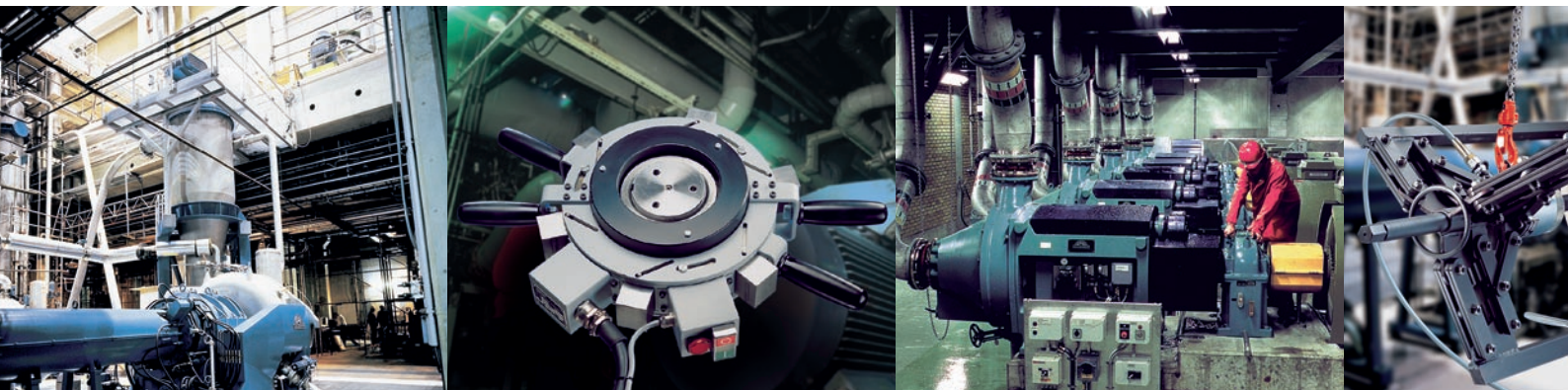
Choosing the right puller for the job is critical. Not only the puller type, but also its maximum withdrawal capacity is crucial for completing any dismounting job safely and easily. Puller overload can result in breakage of the puller's arms and/or beam and therefore should be avoided. This breakage can damage the bearing or shaft and can cause personal injury. In general, it is recommended to use a three-arm puller rather than a two-arm one as the three-arm puller is more stable. Whenever possible, apply the withdrawal force to the ring with the interference fit. SKF offers a complete range of easy-to-use mechanical, hydraulic and hydraulically assisted bearing pullers for use in many bearing applications.

Dismounting using heat

The inner rings of cylindrical roller bearings generally have a tight interference fit, which requires high forces to dismount. In such cases, using a puller can cause damage to the shaft and ring, and can be hazardous to the operator. Using heating equipment facilitates easy and quick dismounting while reducing the risk of damage to the ring and shaft. SKF offers a range of heating equipment, which includes aluminium heating rings as well as adjustable and fixed induction heaters, for dismounting cylindrical roller bearing inner rings.

Dismounting bearings using hydraulic techniques

The SKF hydraulic techniques are often the preferred method for dismounting larger bearings as well as other components. These techniques, which employ hydraulic pumps, nuts and oil injectors, allow the application of substantial forces to dismount bearings or other components.



Safety

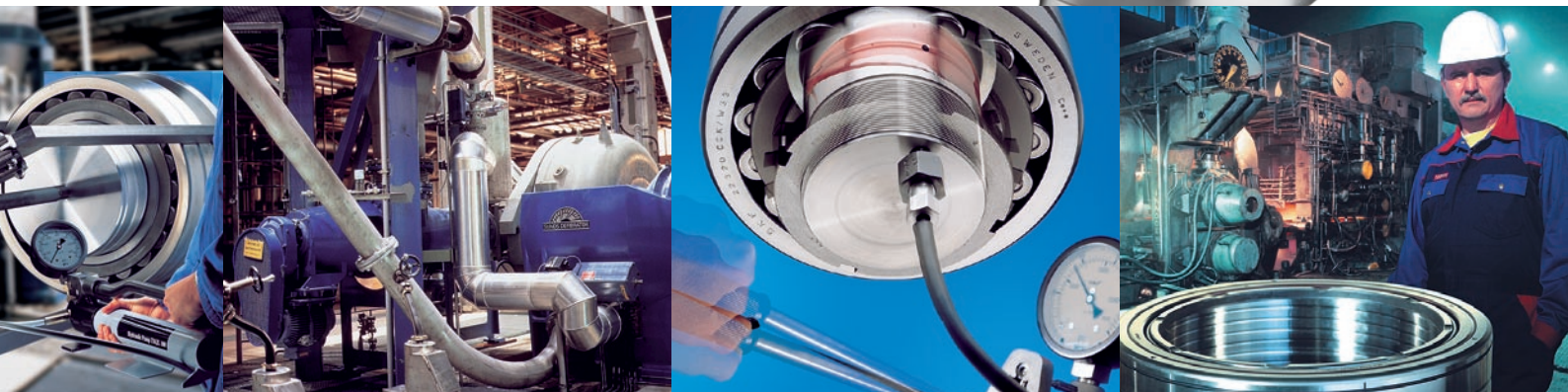
For optimum safety when dismantling bearings:

- Always wear protective clothing and goggles when dismantling bearings
- When dismantling bearings using pullers, make sure to select a suitable puller for the application with sufficient pulling force in order to reduce the risk of puller overload. Overloading a puller can result in puller arm or spindle breakage, causing injury to the operator
- A safety blanket fitted around the puller and bearing helps reduce the risk of injury in case the bearing, puller's arm or spindle break
- Always use heat resistant gloves when dismantling bearings using heat
- For your own safety, do not strike the bearing directly with any hard object such as a hammer or chisel
- Apply dismantling force to part of bearing that has highest fit
- Failure to follow the safety precautions on this page and elsewhere in this section can result in serious injury and damage to the bearings

Mechanical dismantling

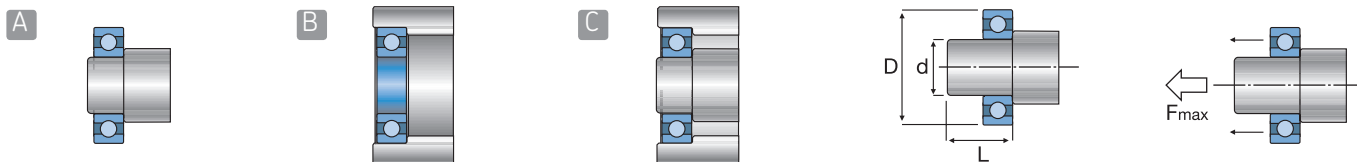
Always the right puller for the job

SKF offers a wide range of bearing pullers for various types of pulls: External, internal and even blind. The range includes mechanical, hydraulic and hydraulically assisted pullers for easy application of high withdrawal forces.





Dismounting



Guide to SKF puller selection

Type of pull	Designation	No. of arms	Width of grip		Effective arm length		Maximum withdrawal force			
			D		L		F max			
			mm	in	mm	in	kN	ton (US)		
EXTERNAL PULL	TMMP 2x65	2	15 – 65	0,6 – 2,6	60	2,4	6	0,7		
	TMMP 2x170	2	25 – 170	1,0 – 6,7	135	5,3	18	2,0		
	TMMP 3x185	3	40 – 185	1,6 – 7,3	135	5,3	24	2,7		
	TMMP 3x230	3	40 – 230	1,6 – 9,0	210	8,3	34	3,8		
	TMMP 3x300	3	45 – 300	1,8 – 11,8	240	9,4	50	5,6		
	TMMP 6	3	50 – 127	2,0 – 5,0	120	4,7	60	6,7		
	TMMP 10	3	100 – 223	3,9 – 8,7	207	8,2	100	11,2		
	TMMP 15	3	140 – 326	5,5 – 12,8	340	13,4	150	17,0		
	TMMA 60	3	36 – 150	1,4 – 5,9	150	5,9	60	6,7		
	TMMA 75H	3	52 – 200	2,0 – 7,8	200	7,8	75	8,4		
	TMMA 80	3	52 – 200	2,0 – 7,8	200	7,8	80	9,0		
	TMMA 100H + .../SET	3	75 – 250	3,0 – 9,8	250	9,8	100	11,2		
	TMMA 120	3	75 – 250	3,0 – 9,8	250	9,8	120	13,5		
	TMHP 10E	3x3	75 – 280	3,0 – 11,0	120 – 200	4,7 – 7,9	100	11,2		
	TMHC 110E	2x3	50 – 170	1,9 – 6,7	70 – 120	2,8 – 4,7	100	11,2		
	TMHP 15/260	3	195 – 386	7,7 – 15,2	264	10,4	150	16,9		
	TMHP 30/170	3	290 – 500	11,4 – 19,7	170	6,7	300	33,7		
	TMHP 30/350	3	290 – 500	11,4 – 19,7	350	13,7	300	33,7		
	TMHP 30/600	3	290 – 500	11,4 – 19,7	600	23,6	300	33,7		
	TMHP 50/140	3	310 – 506	12,2 – 19,9	140	5,5	500	56,2		
	TMHP 50/320	3	310 – 506	12,2 – 19,9	320	12,6	500	56,2		
	TMHP 50/570	3	310 – 506	12,2 – 19,9	570	22,4	500	56,2		
A		No. of extension rod sets	Shaft diameter d		Maximum bearing outer diameter		Effective arm length L		Maximum withdrawal force	
			mm	in	mm	in	mm	in	kN	ton (US)
	TMBS 50E	1	7 – 50	0,3 – 1,9	85	3,3	110	4,3	30	3,4
	TMBS 100E	4	20 – 100	0,8 – 3,9	160	6,3	825 (max)	32,5 (max)	100	11,2
	TMBS 150E	4	35 – 150	1,4 – 5,9	215	8,5	825 (max)	32,5 (max)	100	11,2
	TMHC 110E	2	20 – 100	0,8 – 3,9	160	6,3	255	10	100	11,2
COMBINED INTERNAL OR EXTERNAL PULL *	Designation	No. of extension rod sets	Width of grip D		Width of grip d		Effective arm length L		Maximum withdrawal force	
			mm	in	mm	in	mm	in	kN	ton (US)
	TMMR 40F	2	23 – 48	0,9 – 1,9	59 – 67	2,3 – 2,6	65	2,6	15	1,7
	TMMR 60F	2	23 – 68	0,9 – 2,7	62 – 87	2,4 – 3,4	80	3,2	15	1,7
	TMMR 80F	2	41 – 83	1,6 – 3,3	93 – 97	3,7 – 3,8	94	3,7	30	3,4
	TMMR 120F	2	41 – 124	1,6 – 4,9	93 – 138	3,7 – 5,4	120	4,7	30	3,4
	TMMR 160F	2	68 – 164	2,7 – 6,5	114 – 162	4,5 – 6,4	130	5,1	40	4,5
	TMMR 200F	2	67 – 204	2,6 – 8,0	114 – 204	4,5 – 8,0	155	6,1	40	4,5
	TMMR 250F	2	74 – 254	2,9 – 10,0	132 – 252	5,2 – 9,9	178	7,0	50	5,6
	TMMR 350F	2	74 – 354	2,9 – 13,9	135 – 352	5,3 – 13,9	233	9,2	50	5,6
INTERNAL PULL	Designation	No. of collets	Shaft diameter d		Total sliding hammer length		Weight of hammer			
			mm	in	mm	in	kg	lb		
	TMIP 7-28	4	7 – 28	0,28 – 1,1	412	16,2	1,0	2,2		
B	TMIP 30-60	3	30 – 60	1,2 – 2,4	557	21,9	2,0	4,4		
BLIND PULL *	Designation	No. of arms	Bearing bore diameter d		SKF bearing series		Effective arm length L			
			mm	in			mm	in		
	TMMD 100	6x3	10 – 100	0,4 – 3,9	60..., 62..., 63..., 62/..., 63/..., 64..., 160..., 161...		135 (min.)	5,3 (min.)		
C	TMBP 20E	6x2	30 – 160	1,2 – 6,3	60..., 62..., 63..., 64...		600 (max.)	23,6 (max.)		

* = A bridge construction is needed to support the spindle when used as an internal puller

Guide to SKF puller accessory selection

Puller series	Designation	Safety Puller Protection Blankets TMMX series	Force Generators Advanced Hydraulic Spindle TMHS series	Trisection Pulling Plates TMMS series
TMMP .x.. Standard jaw pullers	TMMP 2x65 TMMP 2x170 TMMP 3x185 TMMP 3x230 TMMP 3x300	– – TMMX 210* TMMX 210 / TMMX 280* TMMX 280 / TMMX 350*	– – – – –	– – TMMS 50* / TMMS 100 TMMS 50* / TMMS 100 TMMS 50 / TMMS 100* / TMMS 160
TMMP .. Heavy duty jaw pullers	TMMP 6 TMMP 10 TMMP 15	TMMX 210 TMMX 280 / TMMX 350 TMMX 350	– – –	TMMS 50* TMMS 100* TMMS 100* / TMMS 160*
TMMR ..F Reversible jaw puller	TMMR 40F TMMR 60F TMMR 80F TMMR 120F TMMR 160F TMMR 200F TMMR 250F TMMR 350F	– – – TMMX 210 TMMX 210 / TMMX 280* TMMX 280* TMMX 350* –	– – – – – – – –	– – – – – – – –
TMMA .. EasyPull	TMMA 60 TMMA 80 TMMA 120 TMMA 75H TMMA 100H TMMA 100H/SET	TMMX 210* / TMMX 280 TMMX 280* / TMMX 350 TMMX 280 / TMMX 350* TMMX 280* / TMMX 350 TMMX 280 / TMMX 350* TMMX 350 **	– TMHS 75 TMHS 100 TMHS 75 ** TMHS 100 ** TMHS 100 **	TMMS 50* TMMS 50* / TMMS 100 TMMS 50 / TMMS 100* / TMMS 160* TMMS 50* / TMMS 100 TMMS 50 / TMMS 100* / TMMS 160* TMMS160 **
TMHC ..E Hydraulic Puller kit	TMHC 110E	TMMX 280* / TMMX 350	TMHS 100 **	TMMS 50 / TMMS 100*
TMHP ..E Hydraulic Puller kit	TMHP 10E	TMMX 210 / TMMX 280* / TMMX 350	TMHS 100 **	TMMS 50* / TMMS 100* / TMMS 160
TMHP .. Hydraulically- assisted heavy duty jaw pullers	TMHP 15/260 TMHP 30/170 TMHP 30/350 TMHP 30/600 TMHP 50/140 TMHP 50/320 TMHP 50/570 TMHP 15/260X TMHP 30/170X TMHP 30/350X TMHP 30/600X TMHP 50/140X TMHP 50/320X TMHP 50/570X	– – – – – – – – – – – – – – – – –	– – – – – – – – – – – – – – – – –	TMMS 160 / TMMS 260 TMMS 260* / TMMS 380 TMMS 260* / TMMS 380 TMMS 260* / TMMS 380 TMMS 260 / TMMS 380* TMMS 260 / TMMS 380* TMMS 260 / TMMS 380* TMMS 160 / TMMS 260 TMMS 260* / TMMS 380 TMMS 260* / TMMS 380 TMMS 260* / TMMS 380 TMMS 260 / TMMS 380* TMMS 260 / TMMS 380* TMMS 260 / TMMS 380*
TMBS ..E Strong back pullers	TMBS 50E TMBS 100E TMBS 150E	TMMX 210 TMMX 210* / TMMX 280 TMMX 280* / TMMX 350	– TMHS 100 ** TMHS 100 **	– – –
TMIP Internal bearing puller kit	TMIP 17-28 TMIP 30-60	– –	– –	– –
TMD 100/TMBP 20E Blind housing puller kits	TMD 100 TMBP 20E	TMMX 210* TMMX 210 / TMMX 280	– –	– –

* = recommended / ** = accessory included with puller



Dismounting



EasyPull: Mechanical Pullers TMM series

Safe and simple bearing dismounting

Equipped with spring-operated arms and a solid design, SKF's patented EasyPull is one of the most user-friendly and safe tools on the market. Ergonomically designed, the spring-operated arms enable the user to position the puller behind the component with just one movement.

- Sturdy design allows dismounting of components even in the tightest application in a safe manner
- The unique red rings spring-operated opening mechanism allows the EasyPull to be placed behind the component with one movement of the hands
- Self-locking arms help prevent the risk of puller slipping under load
- Double hexagonal heads allow easier application of withdrawal force
- Self-centring capability and nosepiece help to avoid damage to shaft
- Efficient use of time due to quick dismounting
- Available in three sizes with a withdrawal force of 60, 80 or 120 kN (6,7, 9,0 or 13,5 ton US), enabling easy selection
- Hydraulic force generators available as an accessory for the 80 and 120 kN versions

The mechanical TMM series consists of 3 (TMM 60, TMM 80 and TMM 120) pullers differing in size and withdrawal force.



EasyPull: Hydraulic Pullers TMM series

Quick and effortless bearing dismounting

The hydraulic versions of the EasyPull, TMM 75H and TMM 100H, combine the user friendliness of the mechanical EasyPull with the effortless force generation provided by integrated hydraulic spindles.

The pullers are protected from overload by safety valves built in their hydraulic spindles.

- Ready-to-use, integrated hydraulic cylinder, pump and puller – thus it is assembly-free and it is not necessary to purchase separate parts
- Safety valve prevents spindles and pullers from being overloaded if excessive force is applied
- The spring-loaded centre point on the hydraulic spindle allows easy centring of the puller on the shaft without damaging the shaft
- The TMM 100H has a maximum withdrawal force of 100 kN (11,2 ton US) and a long stroke of 80 mm (3,1 in), which facilitates most dismounting jobs in just one operation
- For dismounting jobs requiring less force, SKF offers a 75 kN (8,4 ton US) version, the hydraulic EasyPull TMM 75H with a maximum stroke of 75 mm (3 in)
- Supplied with extension pieces and one nosepiece



EasyPull: Hydraulic Puller Set TMMA 100H/SET

A complete bearing dismounting solution

The SKF hydraulic EasyPull set, TMMA 100H/SET, is the most complete dismounting kit available on the market. The set offers the unique combination of the hydraulic EasyPull, a tri-section pulling plate and a puller protection blanket.

In addition to the benefits of the TMMA 100H, which is the essential part of the set, the TMMA 100H/SET also includes:

- A tri-section pulling plate, TMMS 160, that facilitates easy and virtually damage-free dismounting, especially of spherical roller and CARB bearings
- A puller protection blanket, TMMX 350, which is made of transparent material so the user can visually follow the dismounting procedure. It also increases the user's safety while dismounting as it helps to protect from flying fragments of bearings or other components
- A durable metal storage case filled with custom made storage facilities for all parts, minimizing the risk of losing or damaging any of the components
- A complete solution for effortless and safe dismounting for many bearing types, especially spherical roller and CARB® bearings, as well other components such as pulleys and flywheels

Combined together, the components of the set facilitate safe and easy dismounting of bearings, such as spherical roller and CARB® bearings, or other components, such as pulleys and flywheels.



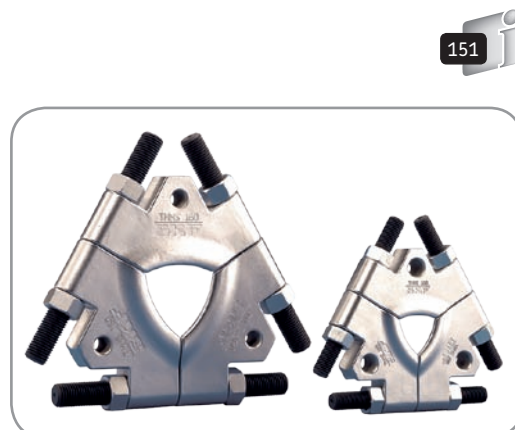
Tri-section Pulling Plates TMMS series

Efficient and correct dismounting

The tri-section pulling plates, TMMS series, are especially designed for use in combination with three-armed pullers. The plates grip behind the inner ring of the bearing. This prevents the pulling forces from being transmitted through the outer ring and the rolling elements,

- The firm grip behind the bearing's inner ring helps ensure that the pulling forces are only transmitted through the inner ring and not through the outer ring or the rolling elements, preventing bearing damage
- The tri-section construction allows even force distribution, preventing bearing locking and/or tilting on the shaft, especially in the case of spherical roller and CARB® bearings, during dismounting
- Suitable for use in combination with three-armed pullers
- Special wedge shape design allows the plates to be easily inserted between the bearing and the shoulder on the shaft
- Available in a wide range, covering most common shaft sizes

minimising the risk of bearing damage. The TMMS series consists of 5 different sizes of tri-section pulling plates suitable for shafts with maximum diameter ranging from 50 to 380 mm (2 to 15 in).





Dismounting



Advanced Hydraulic Spindles TMHS 75 and TMHS 100

Effortless withdrawal force generation

The SKF advanced hydraulic spindles TMHS 75 and TMHS 100 generate a high pulling force with very little effort compared to the standard mechanical spindles. They dramatically reduce the time needed to dismount a bearing or other component.

- Integrated hydraulic cylinder, pump and spindle – no separate pump is required
- Safety valve helps prevent overloading the spindle and the puller in case excessive force is applied
- Spring-loaded centre point in the nosepiece allows easy centring of the puller on the shaft without damaging shaft centre point
- Hand lever with ergonomic grip can be rotated 360°
- Hardened and chrome plated piston with spring return function
- Extension pieces included allows easy adaptation for the required pulling length

The spindles are equipped with integrated hydraulic pumps for force generation. Maximum pulling forces are limited by special safety valves and the hydraulic oil will remain inside the pump.

TMHS 75:

- The TMHS 75 has a maximum withdrawal force of 75 kN (8,4 ton (US)) and a long stroke length of 75 mm (3,0 in), which allows dismounting in one operation
- Suitable for use with any puller equipped with a UN 1 $\frac{1}{4}$ " × 12 tpi threading that can be used up to the maximum force of 75 kN (8,4 ton (US))
- Delivered with a 50 (2,0 in) and 100 mm (3,9 in) extension piece

TMHS 100:

- The TMHS 100 has a maximum withdrawal force of 100 kN (11,2 ton (US)) and a long stroke of 80 mm (3,1 in), which allows dismounting in one operation
- Suitable for use with any puller equipped with a UN 1 $\frac{1}{2}$ " × 16 tpi threading that can be used up to the maximum force of 100 kN (11,2 ton (US))
- Delivered with a 50 (2,0 in), 100 mm (3,9 in) and 150 mm (5,9 in) extension piece

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SKF force generator selection

Puller	Mechanical spindle	TMHS 75	TMHS 100
TMMA 60	■		
TMMA 80	■		
TMMA 120	■	●	●
TMMA 75H		■	
TMMA 100H			■
TMMA 100H/SET			■
TMBS 50E	■		
TMBS 100E			■
TMBS 150E			■
TMHC 110E			■
TMHP 10E			■

■ = Standard with puller

● = Accessory for puller



Standard Jaw Pullers TMMP series

Versatile two and three arm mechanical pullers

One of the most common ways to dismount small to medium size bearings is to use a basic mechanical puller. Using an SKF puller helps to safeguard against damage caused to the

bearing or to the bearing seating during dismounting. SKF Standard Jaw Pullers offer you easy and safe puller operation.

- Range of five different jaw pullers with two or three arms
- Maximum nominal span from 65 to 300 mm (2,6 to 11,8 in)
- Cone system for automatic centring and secure positioning of arms
- Strong springs keep arms apart for easy operation
- Hardened, high quality carbon steel



Reversible Jaw Puller TMMR F series

Combined internal and external puller

The multipurpose SKF TMMR F Reversible Jaw Pullers are able to grip on both the outside as well as the inside of a component, with equal strength. The TMMR F's are available as a complete set, TMMR 8.

- Both internal and external pulling
- Puller for use in every workshop
- Self-locking arms
- Special safety neck avoids damaging of thread and arms
- Hexagonal head on beam allows rotation of bearing during dismounting
- Gripping range from 23 to 350 mm (0,9 to 13,8 in)
- Also available as complete set on a stand





Dismounting



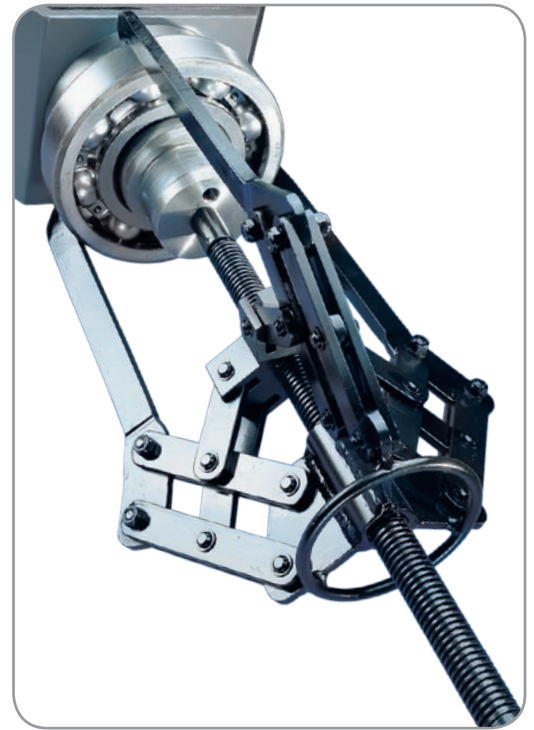
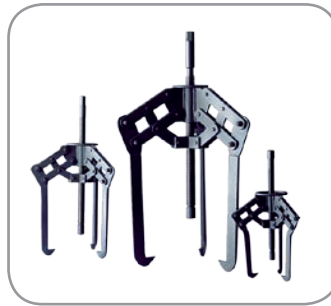
Heavy Duty Jaw Pullers TMMP series

Powerful self-centring mechanical pullers

The SKF TMMP Heavy Duty Jaw Pullers provide perfect alignment and shaft protection as well as an exceptional grip for medium to large size bearings.

- 3 arm jaw pullers with a maximum withdrawal force of 60 to 150 kN (6,7 to 17,0 ton US)
- Unique pantograph system for grip width adjustment that counteracts misalignment during operation
- Fast, efficient and smooth handling
- Blackened, high quality steel for corrosion resistance

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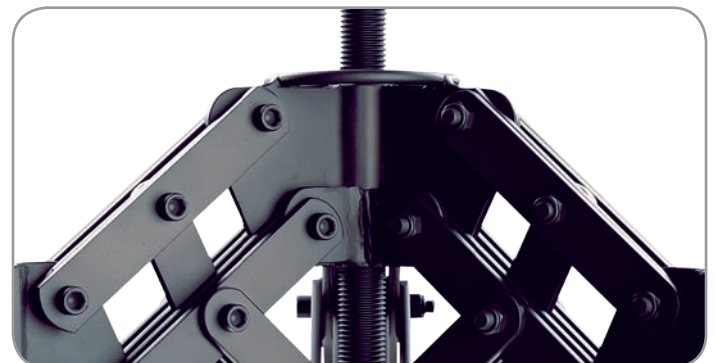
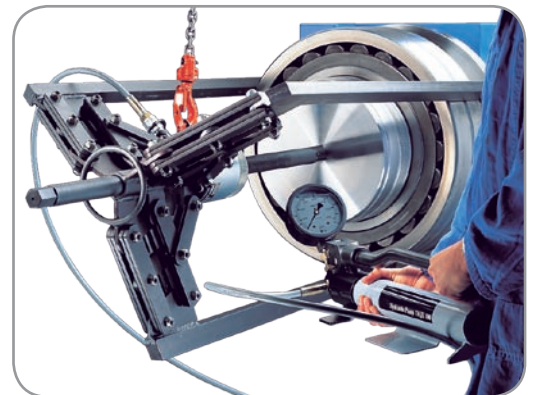
Hydraulically Assisted Heavy Duty Jaw Pullers TMHP series

Powerful self-centring hydraulic pullers

The SKF Hydraulically Assisted Jaw Pullers TMHP 15, TMHP 30 and TMHP 50 are both safe and powerful. They also have a self-centring ability, which is necessary when applying large forces.

- Hydraulically assisted pullers with a maximum withdrawal force of 150, 300 or 500 kN (17,0, 34,0 or 56,0 ton US)
- Pantograph system for ultimate alignment of withdrawal forces and minimised risk of damaging shaft or bearing
- Extreme forces can be applied with ease (self-centring, lifting handle and eye bolt)
- The combination of a spindle and hydraulic cylinder allows the working length to be easily adjusted
- Can be supplied with or without the SKF Hydraulic Pump TMJL 100

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Hydraulic Jaw Puller Kit TMHP 10E

Effortless bearing dismounting up to 100 kN

The SKF TMHP 10E Hydraulic Jaw Puller Kit is equipped with a hydraulic spindle, which facilitates effortless bearing dismounting up to 100 kN (11,2 ton US).

- High load rating of 100 kN (11,2 ton US) makes the puller suitable for a variety of dismounting jobs
- The 3 different arm sizes, with a maximum effective arm length of 200 mm (7,9 in), make the TMHP 10E suitable for use in a wide range of applications
- The self-locking arms minimise the risk of the puller slipping from the application when under load
- Hydraulic spindle facilitates effortless dismounting
- The hydraulic spindle is equipped with a safety valve, which minimises the risk of puller overload by limiting the applied force to 100 kN (11,2 ton US)
- Long stroke of hydraulic spindle, 80 mm (3,1 in), facilitates dismounting in one operation

The versatile puller kit includes three different arm sizes and can be assembled as a two-arm puller or a three-arm puller depending on the space and demand of the application.

- Extension pieces of the hydraulic spindle allow quick adaptation to pulling length
- The spring-loaded centre point of the hydraulic spindle allows easy centring of the puller on the shaft without damaging it



Hydraulic Puller Kit TMHC 110E

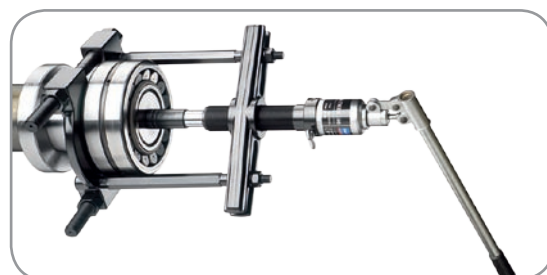
Powerful combination of a jaw and strong back puller

The SKF TMHC 110E is a hydraulic puller kit, which combines a jaw puller and a strong back puller with up to 100 kN

(11,2 ton US) pulling power. The versatile puller kit facilitates safe and easy dismounting in a variety of applications.

- Unique combination of a hydraulic jaw and a strong back puller for use in various applications
- High load rating of 100 kN (11,2 ton US) make the TMHC 110E suitable for use in many applications
- Hydraulic spindle facilitates easy and quick dismounting, effortless generation of force
- The jaw puller includes 2 different arm sizes for maximum reach of 120 mm (4,7 in)
- The jaw puller can be assembled as a three-arm or two-arm puller depending on the space and demands of the application

- The firm grip of the strong back puller behind the bearing's inner ring reduces the force required to dismount the bearing
- The special separator design of the strong back puller allows the puller to be used even in the tightest spaces
- The extension rods of the strong back puller allow a maximum reach of 255 mm (10 in) for quick adaptation to required pulling length





Dismounting



Strong Back Pullers TMBS E series

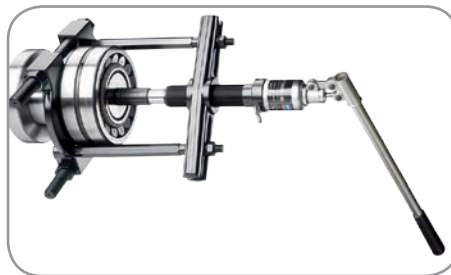
Easy bearing dismounting even in the tightest spaces

The SKF TMBS E strong back pullers facilitate dismounting of bearings in applications where the use of traditional jaw pullers is restricted due to lack of space or where the application demands a long reach.

TMBS 100E and the TMBS 150E are equipped with a hydraulic spindle, which allows for effortless application of force up to 100 kN (11,2 ton US). The TMBS 50E is equipped with a mechanical spindle for force generation.

- Special separator design allows the puller to be easily inserted between the bearing and the shoulder on the shaft, even in the tightest spaces
- The firm grip behind the bearing's inner ring reduces the force required to dismount the bearing
- High load rating of 100 kN (11,2 ton US) make the TMBS 100E and TMBS 150E suitable for use in many applications
- TMBS 100E and TMBS 150E offer complete hydraulic puller kits, allowing effortless generation of force
- The extension rods of the TMBS 100E and TMBS 150E allow easy and quick adaptation to required pulling length
- Maximum reach of 825 mm (32,5 in) and maximum shaft diameter of 150 mm (6 in) allow the TMBS E series to be used in many applications
- The hydraulic spindle is equipped with a safety valve, which limits the applied force to 100 kN (11,2 ton US), minimising the risk of puller overload
- The spring-loaded centre point of the hydraulic spindle allows easy centring of the puller on the shaft without damaging it
- Long stroke of hydraulic spindle, 80 mm (3,1 in), facilitates dismounting in one operation
- Pumping mechanism used for force generation is effortless and more efficient than turning mechanism
- Extension pieces of the hydraulic spindle allow quick adaptation to pulling length

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TMBS 50E



TMBS 100E



TMBS 150E



Deep Groove Ball Bearing Puller Kit TMMD 100

Easy dismounting of bearings in blind housings

The SKF TMMD 100 is especially designed to allow easy and quick dismounting of deep groove ball bearings with interference fit on both rings. The puller is suitable for use in both blind housings and shaft applications. The puller kit contains six sets of different size puller arms,

- The claws are especially designed to facilitate a precise fit in the bearing's raceways, providing exceptional grip and allowing the application of higher dismounting forces
- Each puller arm is fitted with a spring for easy installation
- The puller arms are made of a single piece, laser-cut hardened-steel for strength and longevity
- The hexagon head of the spindle is equipped with an anti-slip pin, which prevents the spanner from sliding down the spindle during dismounting
- The rotation angle of the claw has been limited to allow easy insertion
- The designation is laser-engraved on the arms allowing easy identification and selection
- In addition to dismounting open bearings, the TMMD 100 can be used to dismount sealed bearings after the removal of the seal

two spindles and one handle packed in a carrying case.

The TMMD 100 is suitable for dismounting up to 71 different SKF Deep Groove Ball Bearings from shaft diameter ranging between 10 and 100 mm (0,4 – 3,9 in).



- A** Bearing selection chart included
- B** The rubber cap allows easy and quick attachment of the arms to the spindle. It also prevents the puller arms from detaching from the spindle during operation
- C** The springs are colour-coded for easy arm selection and matching



Selection table

DGBB bearing series	Spindle TMMD 100-S1				Spindle TMMD 100-S2							
	TMMD 100 A1	TMMD 100 A2	TMMD 100 A3	TMMD 100 A4	TMMD 100 A5				TMMD 100 A6			
60..	6000 6001 6002	6004 6005 6006	6007 6008 6009	6011 6012 6013		6014 6015	6016 6017	6018 6019 6020				
62..	6200	6201 6202 6203	6204 6205	6206		6207 6208	6209 6210 6211		6212 6213	6214 6215 6216 6217	6218	
63..		6300	6301 6302	6303 6304		6305	6306 6307	6308		6309 6310	6311 6312	6313
62/ 63/ 64..			62/22	62/28	63/22	63/28	6403	6404 6405	6406	6407	6408 6409	6410
160..	16002 16003		16011									
161..	16100 16101											



Dismounting



Internal Bearing Puller Kits TMIP series

Fast and easy bearing dismounting



The SKF TMIP Internal Bearing Puller Kits are specially designed for dismounting bearings from housings where the fit is on the outer ring. SKF designed, the pullers generate

optimum puller force with use of spring-operated extractors and a unique designed sliding hammer, with no need for additional tools.

- Easy removal of bearings from housings, decreases the amount of time to dismount the bearing
- Specifically designed to suit bearing bore diameters, making the selection of extractors easy
- High-strength chrome coated engineering steel used in the puller construction provides optimum strength and durability
- Spring operated extractors helps result in easy expansion of the extractor to fit the inner ring
- Optimised claw design provides a secure grip behind the inner ring, generating higher puller force with less effort
- Ergonomic sliding hammer facilitates ease of use
- SKF designed, patent pending



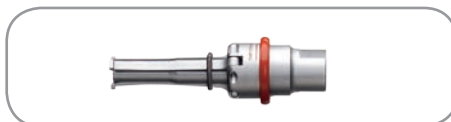
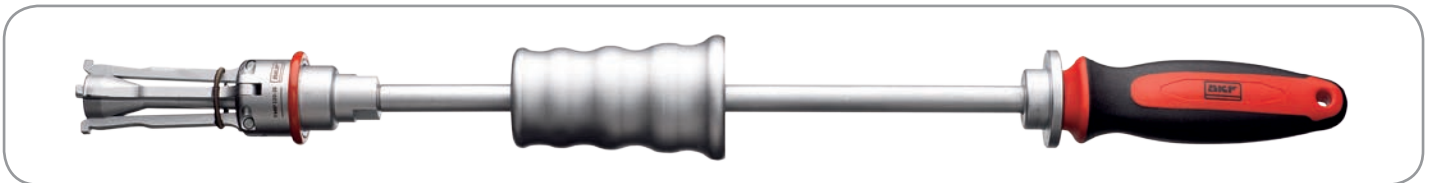
TMIP 7-28

Supplied with 4 extractors covering bore diameters from 7 to 28 mm (0,28 to 1,1 in)



TMIP 30-60

Supplied with 2 extractors covering bore diameters from 30 to 60 mm (1,2 to 2,4 in)



Puller Protection Blankets TMMX series

For safety during dismounting

The SKF Puller Protection Blankets TMMX series, are especially designed to offer additional safety during the dismounting bearings or other components. The TMMX series blankets are simply wrapped around the puller and application after the puller has been positioned.

- Offers additional protection for the user during dismounting jobs
- The tough, transparent plastic allows the user to monitor the component and the puller during operation
- Suitable for use in combination with many pullers
- Especially designed to fit SKF pullers TMMX series

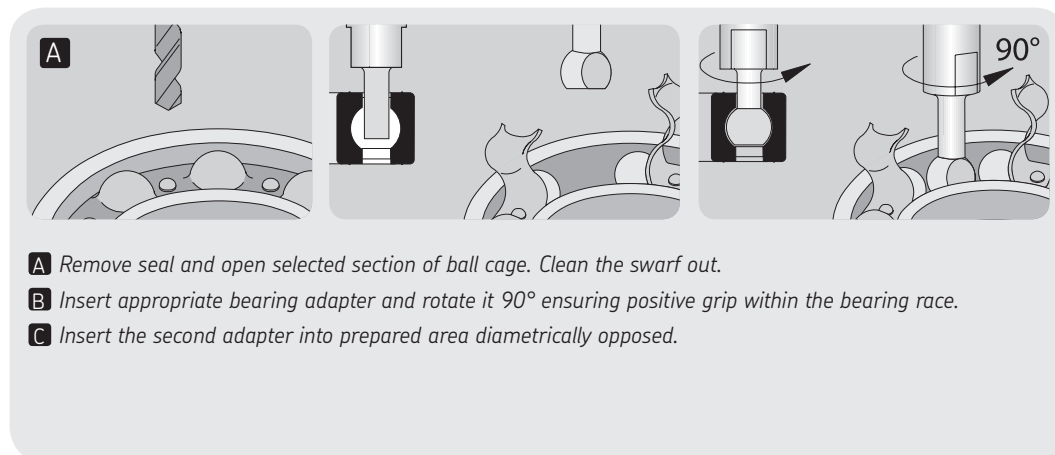


Blind Housing Puller Kit TMBP 20E

Removes bearing without dismantling machinery

The SKF TMBP 20E is an adapter type puller for dismounting deep groove ball bearings in blind housings (i.e. when accessing the ball bearing is not possible through the outer ring or through the bore) with shaft dimensions between 30 mm and 160 mm (1,18 in) and 6,3 in). The use of extension rods allows a long reach of up to 583 mm (23 in).

- With 6 sets of adaptors, a wide of range of Deep Groove Ball Bearings can be dismounted
- New ball adapters are more durable
- Spanner stop function on spindle for easy and safe handling
- Special nose piece helps in minimizing damage to shaft and improves puller stability
- Nosepiece with self-locking function





Dismounting



Combi Kit TMMK 10-35

Multi-purpose kit for quick and easy mounting and dismounting

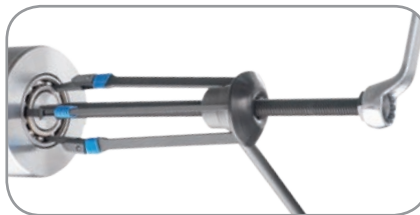
NEW

The SKF Combi Kit TMMK 10-35 is designed for quick and precise mounting of bearings with bore diameters from 10 to 35 mm and for dismounting deep groove ball bearings from shafts, housings and blind housings from the same range.

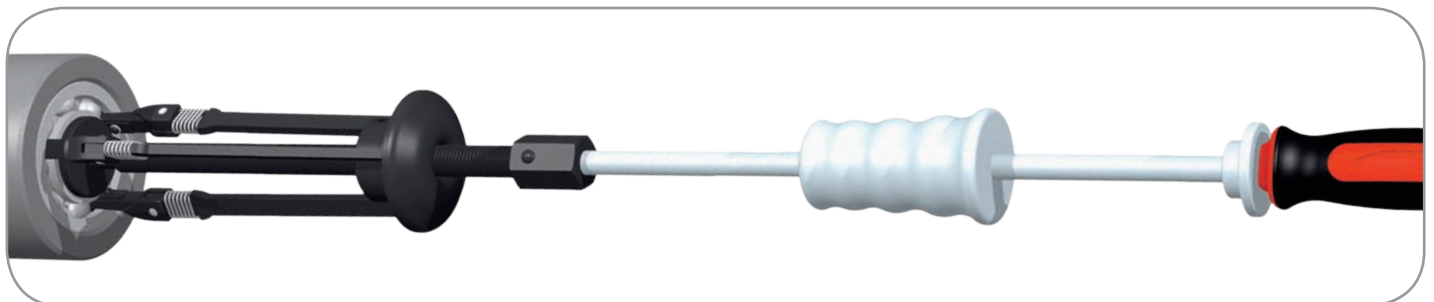
For mounting, a multipurpose fitting tool is included which is suitable for mounting bearings as well as bushings, seal rings, belt pulleys and other similar products.

For dismounting of deep groove ball bearings from blind housings and shafts, the SKF Combi kit contains a unique three armed puller. When dismounting deep groove ball bearings from housings, a combination of this puller, sliding hammer and support rings enables easy removal of bearings.

- All parts are clearly arranged in the case for easy selection and identification
- The correct combination of impact ring and sleeve helps ensure that mounting forces are not transmitted via the rolling elements of the bearing, minimizing damage to the bearing due to incorrect mounting
- The impact rings are made of high-impact modified polyamide. The impact sleeves are made of glass fibre-reinforced, high-impact modified polyamide, which is super-tough, strong and lightweight
- The dead-blow hammer has nylon faces and is steel-shot loaded for maximum impact. The handle, with comfortable rubber-grip for good handling, absorbs shock and vibration
- A complete kit of different sizes of puller arms and spindles facilitates the dismounting of up to 49 different SKF Deep Groove Ball Bearings
- The claws are specially designed to facilitate a precise fit in the bearing's raceways, providing good grip and allowing the application of higher dismounting forces
- The designation is laser-engraved on the arms allowing easy identification and selection
- The springs are colour-coded allowing easy selection and matching
- Elastic locking ring results in easy connection of puller arms to spindle
- Heavy sliding weight of the sliding hammer generates a high dismounting force



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Other dismantling tools

SKF also offers a wide range of mechanical tools, which facilitate dismantling. For more details on these tools,

please see pages 13 – 15 of the mounting and lubrication section of this catalogue.

Index other mechanical dismantling tools		
Designation	Description	Page
HN series	Hook spanners	13
HNA series	Adjustable hook spanners	13
HN ../SNL series	Hook spanners for SNL housings	14
TMFN series	Impact spanners	13
TMFS series	Axial lock nut sockets	15

Dismounting using heat

Easy, quick and safe dismantling of cylindrical roller bearing inner rings

SKF's range of heating equipment enables quick and safe dismantling of cylindrical roller bearing inner rings and covers a wide range of applications. Aluminium heating rings TMBR series are designed for dismantling inner rings of small and medium-size cylindrical roller bearings.

Adjustable and fixed induction heaters EAZ series are suitable for frequent dismantling of various sizes of cylindrical roller bearing inner rings.

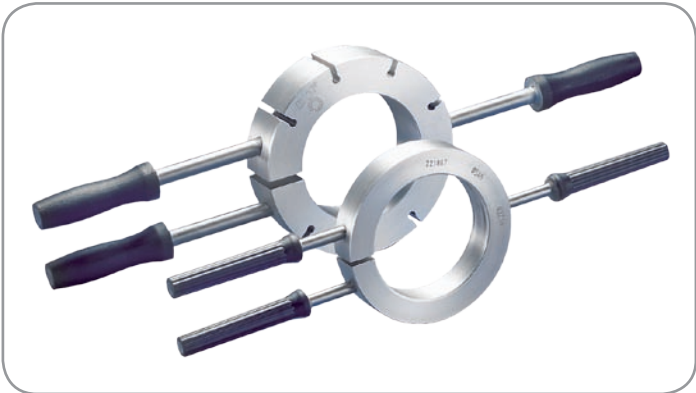
Aluminium Heating Rings TMBR series

For regular dismantling of cylindrical roller bearings

The aluminium heating rings are designed for dismantling inner rings of cylindrical roller bearings. They are available for all bearing sizes of the NU, NJ and NUP series, these are

bearings without flanges or with only one flange on the inner ring. The rings are available as standard for the following bearing sizes: 204 to 252, 304 to 340, 406 to 430.

- Simple and easy-to-use
- Avoids shaft and bearing inner ring amage





Dismounting



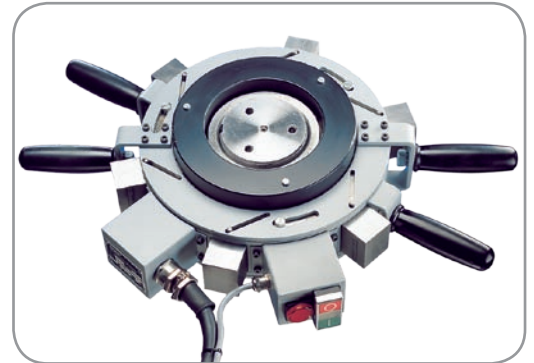
Adjustable Induction Heaters EAZ series

For frequent dismounting of cylindrical roller bearings

The SKF Adjustable Induction Heaters EAZ 80/130 and EAZ 130/170 are used for frequent dismounting of cylindrical bearing inner rings. Where inner rings are removed infrequently, aluminium heating rings, TMBR series, are also available.

- Covers most cylindrical bearings 65 to 130 mm (2,5 to 5,1 in) bore diameter
- Wide range of power supplies
- 1 year warranty
- Avoids shaft and bearing inner ring damage
- Fast and reliable bearing removal
- Up to n6 interference fit

For larger cylindrical inner rings normally found in steel mill applications, SKF can supply special EAZ induction heaters.



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Selection table for bearings NJ-NUP

Designation

EAZ 80/130	213-220	313-319	412-417	1014-1022	2213-2220	2313-2319
EAZ 130/170	222-228	321-324	419-422	1024-1030	2222-2228	2322-2324

All E-types included.

Selection table for bearings NU

Designation

EAZ 80/130	213-221	313-320	412-418	1014-1022	2213-2220	2313-2320
EAZ 130/170	222-228	321-326	419-424	1024-1030	2222-2228	2322-2326

All E-types included.

Fixed Induction Heaters EAZ series

Quick roll changes with bearing removal in 3 minutes

In light section mills and wire rod mills, four-row cylindrical roller bearings are usually used to take up the roll separating forces. The inner rings of these bearings are mounted with an interference fit on the roll necks.

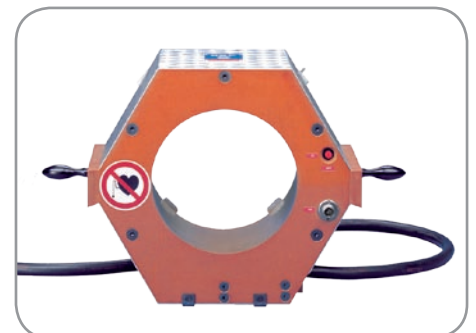
Because of the rapid wear, heavy loads and severe contamination the rolls must be frequently replaced. This inevitably involves dismounting the inner rings and remounting them onto new rolls.

Three minutes are enough

Using the EAZ, inner rings are heated evenly while the roll neck remains cold. The ring, together with the induction heater, can be easily withdrawn from the neck. Even with relatively large rings this complete operation takes not more than two to three minutes.

- Reduced time to remove bearings
- Increased production time
- Available in different voltage versions
- Bearings can be reused
- Control cabinet is to be ordered separately

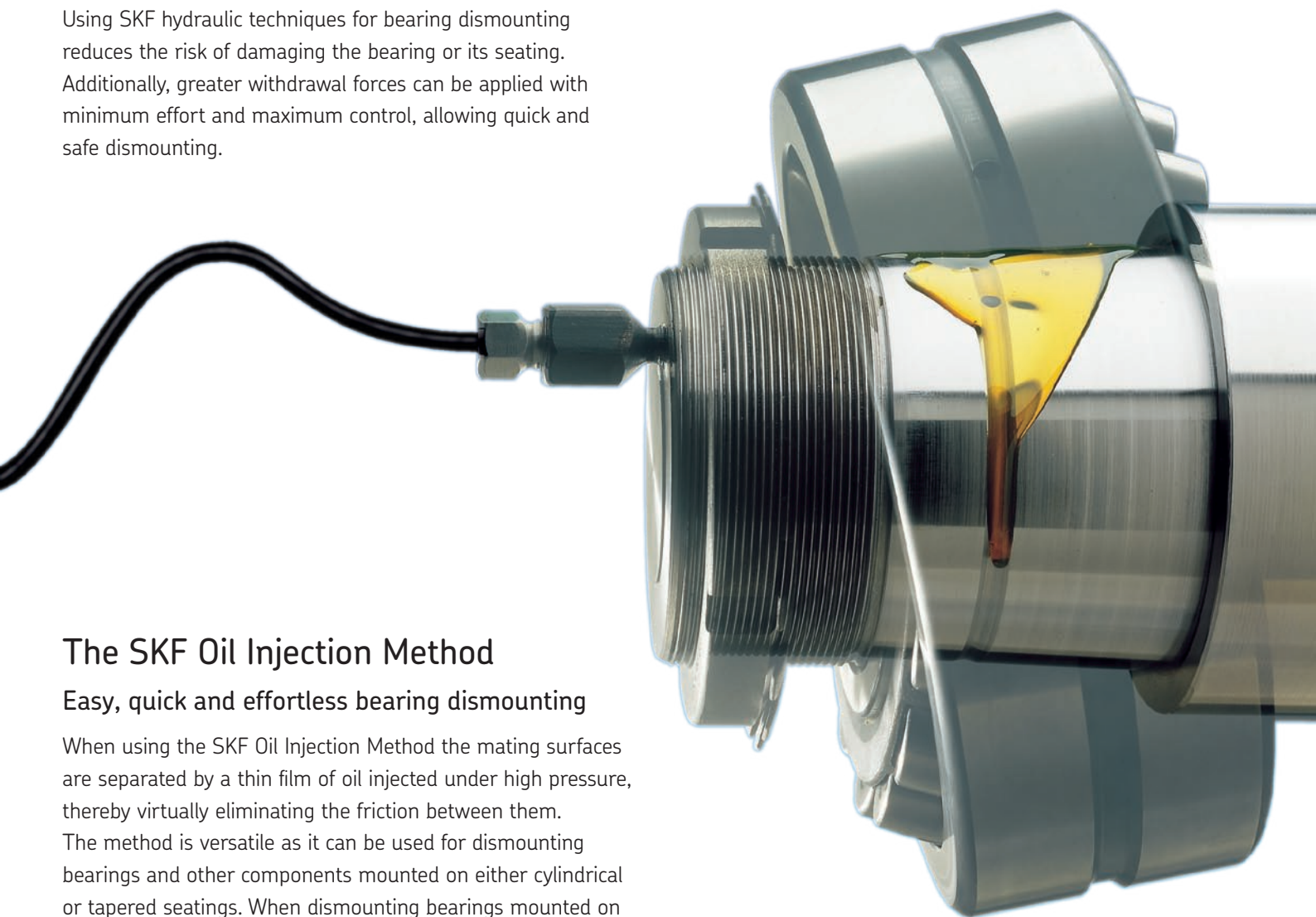
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Dismounting bearings using hydraulic techniques

Correct and quick bearing dismounting

Using SKF hydraulic techniques for bearing dismounting reduces the risk of damaging the bearing or its seating. Additionally, greater withdrawal forces can be applied with minimum effort and maximum control, allowing quick and safe dismounting.



The SKF Oil Injection Method

Easy, quick and effortless bearing dismounting

When using the SKF Oil Injection Method the mating surfaces are separated by a thin film of oil injected under high pressure, thereby virtually eliminating the friction between them. The method is versatile as it can be used for dismounting bearings and other components mounted on either cylindrical or tapered seatings. When dismounting bearings mounted on cylindrical seatings, the injected oil can reduce the required pulling forces by up to 90%.

Subsequently, the physical effort required when using a puller to remove the bearing from its seating is reduced. When using the Oil Injection Method to dismount bearings mounted on tapered seatings, the interference fit is completely overcome by the injected oil. The bearing is then ejected from the seating with great force, making the use of a puller unnecessary. In this case, a stop-nut must be used to control the ejection of the bearing.

The method, which is used for many bearing applications, can also be found in other applications, such as:

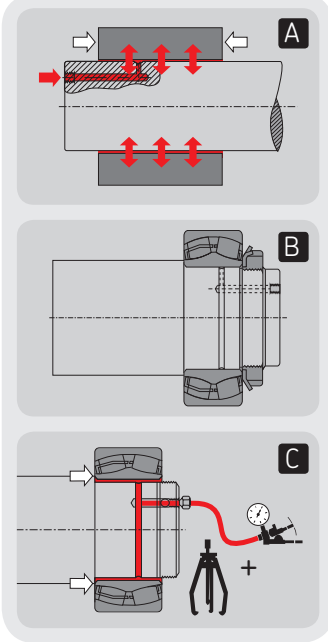
- Couplings
- Gear wheels
- Railway wheels
- Propellers
- Built-up crankshafts



Dismounting



Cylindrical shafts



The concept

A By injecting oil of a certain viscosity between two shrink fitted surfaces, the mating surfaces will be separated by a thin oil film. The dismounting force required is thus greatly reduced. The thin oil film also minimises the risk of metallic contact when dismounting, reducing the risk of component damage.

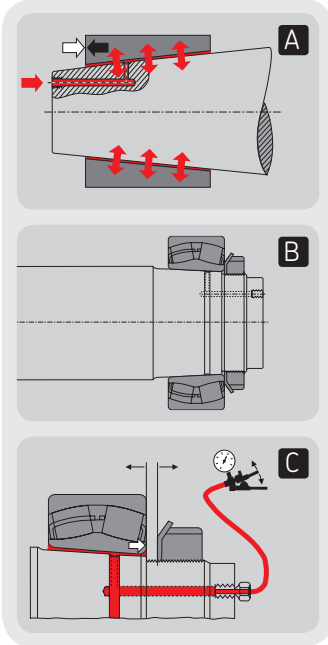
The preparation

B During manufacture the shafts are prepared with oil ducts and grooves. For technical information on how to prepare the shafts, consult an SKF application engineer.

The action

C Dismounting the bearing is made easy by pumping oil under pressure between the mating surfaces. Once the oil pressure has built up, the component can be removed from the shaft with a minimum of effort.

Tapered shafts



The concept

A Injecting the oil between two tapered surfaces will create a reaction force which could be quite substantial as the oil will also act as a "hydraulic cylinder" which can push the outer component off.

The preparation

B During manufacture the shafts are prepared with oil ducts and grooves. For technical information on how to prepare the shafts, consult an SKF application engineer.

The action

C Bearings are dismounted by injecting oil between the mating surfaces and when sufficient pressure is reached, the bearing will be pushed off. A nut is required to keep the bearing from sliding off the shaft.



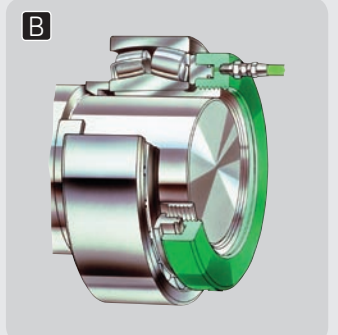
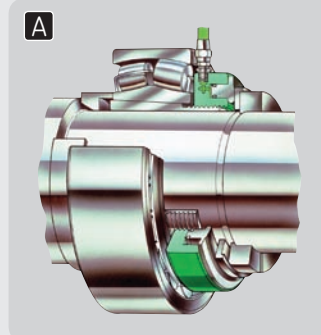
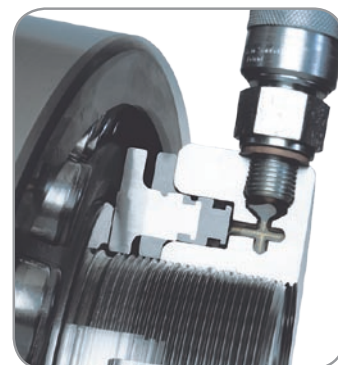
In addition to dismounting bearings mounted on cylindrical or tapered seatings, the SKF Oil Injection Method can be also used for mounting bearings on tapered seatings. See page 22-23 of this catalogue for more details.



Hydraulic Nuts HMV ..E series

Effortless dismounting of bearings mounted on sleeves

Dismounting bearings mounted on either adapter or withdrawal sleeves is often a difficult and time-consuming job. These problems can be reduced with the use of an SKF hydraulic nut. Oil is pumped into the nut and the piston is pushed out with a force, which is sufficient to free the sleeve. All HMV ..E nuts are supplied with a quick connection coupling to fit the SKF hydraulic pumps.



- A** HMV ..E nut and stop ring in position to press an adapter sleeve free.
- B** HMV ..E nut used to free a withdrawal sleeve.



SKF hydraulic nuts HMV ..E series also facilitate bearing mounting. For more details see page 26 of this catalogue.

Dismounting Fluid LHDF 900

For easy and quick bearing dismounting

The Dismounting Fluid LHDF 900 is suitable for use with SKF hydraulic equipment, including hydraulic pumps and oil injection tools. The LHDF 900 contains anti corrosives which are non aggressive to seal materials such as nitrile rubber, perbunan, leather and chrome leather, PTFE, and so on.

Ordering details and technical data

Designation	LHDF 900/pack size
Specific gravity	0,885
Flash point	202 °C (395 °F)
Pour point	-28 °C (-18 °F)
Viscosity at 20 °C (68 °F)	910 mm ² /s
Viscosity at 40 °C (104 °F)	330 mm ² /s
Viscosity at 100 °C (212 °F)	43 mm ² /s
Viscosity index	180
Available pack size	5 and 205 litre

